CLAD: Robust Audio Deepfake Detection Against Manipulation Attacks with Contrastive Learning

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***Abstract*—The increasing prevalence of audio deepfakes poses significant security threats, necessitating robust detection methods. While existing detection systems exhibit promise, their robustness against malicious audio manipulations remains underexplored. To bridge the gap, we undertake the first comprehensive study of the susceptibility of the most widely adopted audio deepfake detectors to manipulation attacks. Surprisingly, even manipulations like volume control can significantly bypass detection without affecting human perception. To address this, we propose CLAD (Contrastive Learning-based Audio deepfake Detector) to enhance the robustness against manipulation attacks. The key idea is to incorporate contrastive learning to minimize the variations introduced by manipulations, therefore enhancing detection robustness. Additionally, we incorporate a length loss, aiming to improve the detection accuracy by clustering real audios more closely in the feature space. We comprehensively evaluated the most widely adopted audio deepfake detection models and our proposed CLAD against various manipulation attacks. The detection models exhibited vulnerabilities, with FAR rising to 36.69%, 31.23%, and 51.28% under volume control, fading, and noise injection, respectively. CLAD enhanced robustness, reducing the FAR to 0.81% under noise injection and consistently maintaining an FAR below 1.63% across all tests. Our source code and documentation are available in the artifact repository (https://github.com/CLAD23/CLAD).**

***Index Terms*—Audio Deepfake Detection, Manipulation Attacks, Contrastive Learning**

I. Introduction

In recent years, deep learning has made remarkable progress in speech synthesis [1]–[3] and voice conversion [4]–[6]. These technologies advanced the creation of highly realistic and natural-sounding speech, which enhances user interaction in various applications. However, they also pose serious risks. Malicious attackers may use these technologies to generate high-quality audio deepfakes and then exploit them for phone scams [7], bypassing speaker recognition [8] and spreading disinformation on the web [9]. In response, researchers have developed various methods to detect audio deepfakes, including acoustic features such as spectral features [10], linear frequency cepstral coefficients (LFCC) [11] , and constant Q cepstral coefficients (CQCC) [12], as well as deep learning approaches such as RawNet2 [13], Res-TSSDNet [14] and AASIST [15]. These methods have shown impressive performance on large-scale public datasets like the ASVspoof dataset [16], [17].

Despite the impressive performance in detecting audio deepfakes, these methods' effectiveness and robustness in real-world scenarios have yet to be investigated, especially when faced with simple but natural audio manipulations. Most existing methods are evaluated using raw output of synthesis techniques, neglecting the possible manipulations applied to the audio by the adversary [11], [13]–[15]. While the latest ASVspoof 2021 dataset [17] introduces more realistic data subjected to telephony channel transmission and compression, these modifications are significantly limited, and fail to reflect the techniques a real attacker might employ to fool the detection system. Therefore, a thorough understanding of these methods' robustness against simple manipulations is essential to ensure their efficacy in mitigating deepfake threats.

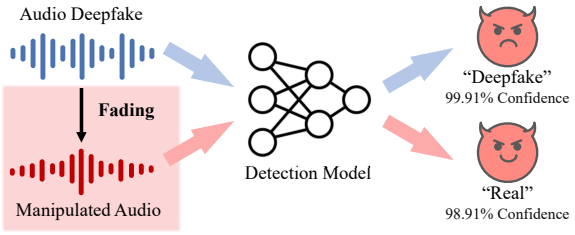


Fig. 1. Illustration of the manipulation attacks. Though detection model performs well on the original audio, we found that simple manipulations like fading could bypass it.

To bridge the gap, we perform the first systematic large-scale study on audio manipulation attacks in the audio deepfake detection task, where an attacker aims to deceive the detection system by simply manipulating the audio data in an imperceptible way. For example, the attacker may apply fading to the audio to bypass the detector, as shown in Fig. 1. There have been previous works [18]–[20] that use adversarial attacks to evade detection systems, these attacks are costly and often require prior knowledge of the target system. In contrast, the manipulation attack is more realistic and cheaper, posing a serious threat to detection systems. Specifically, we design 7 manipulations, i.e., noise injection, volume control, fading, time stretching, resampling, time shifting, echoes adding, and investigate the robustness of widely adopted detection methods against these manipualtions. We observe that these methods fail to counteract the manipulation attacks and the performance drops significantly under these attacks.

In this paper, we propose a Contrastive Learning-based Audio deepfake Detector (CLAD) to enhance the robustness against audio manipulation attacks. The key idea is to use contrastive learning to train a robust audio encoder that effectively differentiates audio deepfakes by learning intrinsic features.

By training the encoder to produce similar feature representations for the same audio under different augmentations, and dissimilar representations for different audios, the encoder can learn robust features that are more resistant to manipulation attacks. To further enhance the effectiveness of CLAD, we developed length loss, which compels the encoder to produce short feature vectors for real audios and long feature vectors for audio deepfakes. By clustering the features of real audios around the origin of the high-dimensional space and pushing the features of deepfakes away, the length loss effectively utilizes label information to improve the detection performance of CLAD. The key insight is that real audios exhibit more resemblance to one another than audio deepfakes, which are generated from distinct synthesis methods.

In summary, our paper makes the following contributions:

- We perform the first comprehensive study on the robustness of widely adopted audio deepfake detection methods against audio manipulation attacks and expose the significant threat of manipulation attacks. We find that these detection methods are vulnerable to simple manipulation attacks, such as volume control and fading.
- To the best of our knowledge, we present the first audio deepfake detection method that is robust to manipulation attacks, namely CLAD. To achieve this, we introduce contrastive learning, which learns a robust encoder to produce reliable features for different audio manipulations. To enhance the accuracy, we also design length loss as a new learning strategy to enhance the encoder.
- We conduct extensive experiments to evaluate the effectiveness of CLAD in defending against manipulation attacks under different scenarios. The results suggest that CLAD is effective and robust against manipulation attacks. e.g., achieving an overall FAR of less than 1.63% across all manipulations. CLAD also shows the significant improvements over existing methods.

II. Manipulation Attacks

A. Manipulations

To attack the audio deepfake detection systems, adversaries attempt to manipulate the audio with simple and common methods. We discuss many kinds of manipulation approaches as never before, some of which have received little attention in prior work [21], [22].

Noise injection applies additive noise to the audio signal, which is a common occurrence in real-world environments. We consider two typical noise types, gaussian white noise (WN) and environmental noise (EN). For our experimental evaluation, we consider the following environmental noises: wind, footsteps, breathing, coughing, rain, clock tick and sneezing. We control the strength of the noise by adjusting the signal-to-noise ratio (SNR).

Volume control (VC) scales the magnitude of a speech signal, while not changing the semantics of the speech signal, but only its loudness. Since human perception of loudness is logarithmic, humans are not sensitive to linear changes in volume. In contrast, the model takes the audio as a sequence of sampling values that are linearly related to the volume.

Fading (FD), including fade in and fade out, adds a smooth transition to the beginning and end of an audio, which makes the audios sound more natural. We apply five different fading shapes to the original signal, namely: linear, logarithmic, exponential, quarter sinusoidal and half sinusoidal.

Time stretching (TS) modifies the speed or duration of an audio signal without affecting its pitch. It is commonly used in audio signal processing to match the tempo of two songs, or to change the duration of an audio signal to fit a fixed length.

Resampling (RS) changes the sampling rate of an audio signal, a technique commonly employed in digital signal processing. This method converts signals from one sampling rate to another, ensuring compatibility between audio files, devices, or to decrease the size of an audio file.

Time shifting (SF) is a process to shift the audio in time domain, which can be used to delay or advance the audio signal by a certain amount of time. For human perception, regardless of how the audio is shifted, the auditory experience remains unchanged. However, for the model, the audio is a sequence of sampling values, the position of the sampling values do impact the model's performance.

Echoes adding (EC) applies echoes to a speech signal by creating delayed and attenuated copies of the original signal and adding them together. Adding echoes makes the speech sound more reverberant, depending on the delay and attenuation factors, while not affecting human recognition of speaker and content.

It is important to note that the manipulations described here are widely employed in audio processing. This simplifies the implementation and ensures that the resulting audio remains perceptually natural to humans. [1] While prior research has considered pitch shift, time masking, and frequency masking [21], [22], we opt not to incorporate these manipulations into our study due to their potential to introduce perceptually unnatural audio artifacts and affect human recognition of speaker identity and content, contradicting the purpose of audio deepfake.

B. Manipulation Attacks vs. Adversarial Attacks

Manipulation attacks are similar to adversarial attacks in that both aim to evade detection systems and achieve malicious purposes. However, manipulation attacks offer three key advantages. First, manipulation attacks are simple and do not require professional knowledge of machine learning. Second, manipulation attacks are computationally efficient. Adversarial attacks is based on optimization, making the generation of successful adversarial examples expensive. In contrast, the manipulations used in our work can be applied quickly to audio deepfakes. The third advantage is that manipulation attacks do not need any prior knowledge of the detection model. Traditional adversarial attacks require full information of the target model, while black box adversarial attacks struggle with transferability issues [20]. We believe these advantages make manipulation attacks more accessible to attackers, posing a serious threat to audio deepfake detection systems.

[1] Our manipulated audio samples are accessible at the following URL: https://sites.google.com/view/clad-demo/.

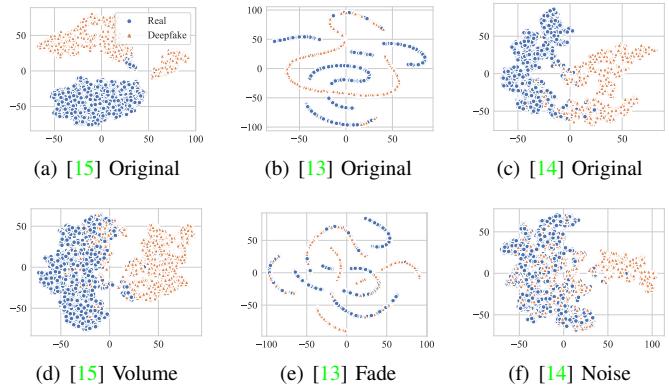


Fig. 2. Visualization of origianl vs. manipulated features extracted by widely adopted detection models via t-SNE. (a) and (d) are features extracted by AASIST, (b) and (e) are features extracted by RawNet2, (c) and (f) are features extracted by Res-TSSDNet.

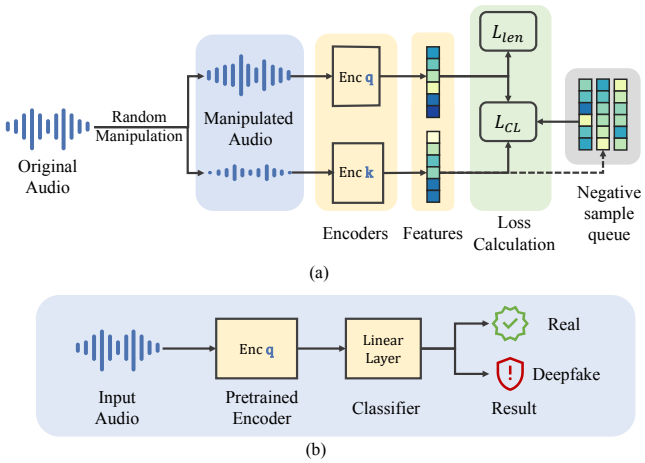


Fig. 3. Overview of CLAD. (a) illustrates the pretraining stage. (b) illustrates the downstream training stage.

III. Motivation

To expose the impact of manipulation attacks, we employ three common techniques: volume control, fading, and noise injection as illustrations to manipulate the audio deepfake samples. Fig. 2 visualizes the feature distribution extracted by three widely adopted models, AASIST [15], RawNet2 [13] and Res-TSSDNet [14] via t-distributed stochastic neighbor embedding (t-SNE). We can observe that compared with original distribution in Fig. 2(a-c), the average distance between manipulated audio deepfakes and real samples is significantly reduced in Fig. 2(d-f). Consequently, this leads to a notable degradation in the performance of the detection models when dealing with manipulated samples.

The underlying cause of this phenomenon is multifaceted. Firstly, existing detection methods have not accounted for the manipulations that could be potentially applied to the data. Secondly, these methods were predominantly trained using conventional supervised learning paradigms, thereby optimized to make predictions based on the specific examples they encountered during training. Manipulations cause the features extracted by these models changed significantly, resulting in the misclassification of the manipulated samples. This observation motivates us to design a robust model capable of withstanding potential manipulations. However, conventional supervised learning approaches often struggle to accommodate such variations due to their inherent limitations in adaptability.

In response to this challenge, we explore the potential of contrastive learning, a promising technique that empowers models to learn invariant and discriminative representations by distinguishing between similar and dissimilar examples based on their features. This enables models to capture the intrinsic characteristics of input audio, even in the presence of diverse manipulations. In contrast, conventional supervised learning approaches often struggle to handle such variations since they are tailored to specific instances and lack the inherent adaptability required. Motivated by these insights, we propose a contrastive learning-based detection model that learns more robust representations.

Furthermore, we observed that that models trained solely on contrastive learning extract features with poor clustering properties, making it challenging to achieve satisfactory performance in the detection task. Given the observation that real samples are more similar to each other than audio deepfakes, we introduce length loss which clusters real audio representations by controlling length of the feature vector. While contrastive learning focuses on the directionality of features, length loss targets their magnitude, thereby not only avoiding interference with the contrastive learning process but also complementing it, boosting the model's performance.

IV. CLAD Design

A. Overview

CLAD is constructed based on the contrastive learning framework presented in [23]. We enhance the framework with additional audio augmentations and length loss. The CLAD model for audio deepfake detection task consists of two main components: an encoder and a linear classifier. The training of the CLAD model is conducted in two phases: pre-training and downstream training, as s illustrated in Fig. 3.

During pre-training, our objective is to train a robust encoder that can produce representative features. Firstly, we randomly apply different manipulations to the input training data, resulting in two manipulated samples. We then feed the manipulated samples into the encoders to obtain the corresponding features. In the context of contrastive learning, samples augmented from the same audio are deemed positive pairs, while those from different audios are considered negative pairs. The encoder is trained to produce similar features for positive pairs and dissimilar features for negative pairs. This training paradigm facilitates the model in learning to generate consistent features for samples, even when subjected to different manipulations, while still maintaining discriminative features across different audios. To enhance the contrastive training, we employ a queue to store previous negative samples, as suggested in [23]. The contrastive loss is computed using the features output by the encoders and negative samples stored in the queue. Additionally, we introduce length loss to aggregate features for

real samples, thereby augmenting downstream performance. Further details regarding the contrastive loss and length loss are elaborated in following sections. The final loss function for pre-training is expressed as the weighted sum of the contrastive loss and length loss, as shown in Eq. 1.

$$\mathcal{L}_{pretrain} = \mathcal{L}_{CL} + \lambda \mathcal{L}_{len} \tag{1}$$

In the downstream training stage, we utilize the pre-trained encoder and add a linear classifier on top of it. We then train the model on the audio deepfake detection task using the labeled dataset to obtain the final detector.

Note that we do not specify the encoder architecture in CLAD. Instead, most deep learning-based end-to-end detection methods can be used as the encoder by removing the classification head. This makes CLAD a plug-and-play solution that can be easily integrated with existing detection methods to improve their robustness.

B. Contrastive Learning

Contrastive learning aims to learn consistent and discriminative representations by training a model to map similar instances closer together and dissimilar instances farther apart in a latent feature space. We employ contrastive learning to train the encoder to produce consistent representations under various manipualtions, enhancing the model's resistance against manipulation attacks.

While prior contrastive learning methods for audio [24], [25] primarily relied on the SimCLR framework [26], we opt for MoCo [23] because it allows for the use of a larger number of negative samples through the maintenance of a negative sample queue. This framework employs two different encoders, named query encoder and key encoder, which are the Enc q and Enc k shown in Fig. 3. The encoders have identical architecture, and the parameters of the key encoder are smoothly updated by using the parameters of the query encoder. [2] They take the different augmentations of input and produce corresponding features, q and k. The augmentations include all the manipulations discussed in Sec. II and are randomly applied during training. We train the encoder using contrastive loss to ensure that the generated feature k is as close as possible to feature q compared to the negative sample features in the queue. This enables the encoder to distinguish between different samples under manipulations. The contrastive loss is defined in Eq. 2.

$$\mathcal{L}_{CL} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp(q_i^T k_i^+/\tau)}{\sum_{j=1}^{K} \exp(q_i^T k_j/\tau)} \tag{2}$$

where N is the batch size, K is the size of the negative sample queue, τ is the temperature parameter, and q_i and k_i^+ are the feature q and feature k of the i-th sample in the batch and k_j is the feature k of the j-th sample in the negative sample queue. After the contrastive loss is computed, we update the parameters of the query encoder with gradient descent. The parameters of the key encoder are updated by using the momentum update rule formulated in Eq. 3.

$$\theta_k = \mu\theta_k + (1-\mu)\theta_q \tag{3}$$

where θ_k and θ_q are the parameters of the key encoder and the query encoder, respectively, and μ is the momentum parameter. Finally, we enqueue the feature k in the batch to the negative sample queue and dequeue the earliest features in the queue to update the queue.

[2]The design of momentum update and negative sample queue aims to increase the negative sample size and stabilize the training process, thereby enhancing the model performance. We refer readers to [23] for more details.

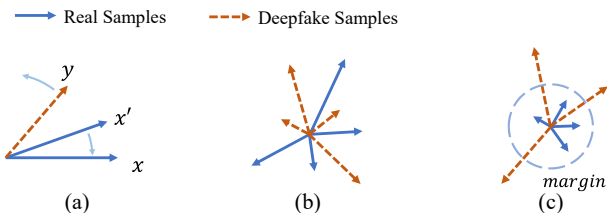


Fig. 4. Illustration of the motivation of length loss. (a) illustrates the training of contrastive loss. (b) illustrates the features extracted by contrastive loss trained encoder. (c) illustrates the features extracted by length loss and contrastive loss trained encoder.

C. Length Loss

The contrastive loss we used is based on the cosine similarity. However, by solely relying on the cosine similarity for calculating the contrastive loss, the model only learns to minimize the angles between feature vectors corresponding to augmented audios from the same source while increasing the angles for different audios, as depicted in Fig. 4 (a). After such training, we observed that the features extracted by the model remain scattered and exhibit poor clustering, as illustrated in Fig. 4 (b). In the context of audio deepfake detection, real speech samples tend to exhibit higher similarity to one another, whereas synthetic speech, generated using different algorithms, often poses challenges for clustering [27]. Building upon this observation, we propose clustering the feature vectors of real audio to enhance performance. To accomplish this, we introduce length loss that leverages the length information, which is not utilized in the cosine similarity loss calculation. Specifically, we decrease the length of feature vectors for real speech and increase it for audio deepfakes, leading to improved clustering of feature vectors for real speech, as demonstrated in Fig. 4 (c). The length loss is defined mathematically as Eq. 4.

$$\mathcal{L}_{len} = \frac{1}{N}\sum_{i=1}^{N} y_i w \|q_i\|_2 + (1-y_i)\max(margin - \|q_i\|_2, 0) \tag{4}$$

where N is the batch size, q_i denotes the feature of the i-th sample in the batch extracted by encoder $Enc\,q$, y_i is the label of the sample, w is the weight factor to assign different levels of importance to the different classes, since the audio deepfake detection dataset might be highly unbalanced. $margin$ is the margin used to control the degree of separation between the real audios and audio deepfakes.

In summary, length loss is designed to encourage the encoder to generate shorter feature vectors for real audios and longer ones for audio deepfakes. This is motivated by the observation that real speech samples generally exhibit more similarity compared to audio deepfakes generated by various algorithms. Besides, when integrated with the contrastive loss, which focuses on the directionality of feature vectors, the length loss provides an additional dimension of discriminative information. This combination of loss functions allows the model to leverage both magnitude and direction of feature vectors, resulting in improved performance in detection.

D. Downstream Training

After training the encoder with both the contrastive and length loss, we combine the encoder Enc q with a linear layer and train it by minimizing the cross-entropy loss for the downstream task. The linear classifier is a one-layer fully connected neural network that maps the encoder output to the classification score. We define the loss function as follows:

$$\mathcal{L}_{downstream} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(p_i) + (1-y_i)\log(1-p_i)] \quad (5)$$

where N is the batch size, y_i is the label, p_i is the predicted probability of being real for the i-th sample in the batch.

V. Experimental Results

The objective of our evaluation is to answer the following Research Questions(RQs):

- RQ1: How do manipulation attacks degrade the performance of widely adopted audio deepfake detection systems?
- RQ2: How effective is the proposed CLAD model in detecting audio deepfakes under manipulation attacks?
- RQ3: What is the impact on the performance of the CLAD model when contrastive learning or length loss is removed?
- RQ4: How does the performance of the CLAD model change when facing unknown manipulations?

A. Experimental Settings

Baseline Models. We select RawNet2 [13], AASIST [15], Res-TSSDNet [14], and SAMO [28] as baseline models for audio deepfake detection. They are the most widely adopted and influential models in this field, selected as the baseline models for many challenges [17], [29], [30], and have been studied in recent works [31]–[33]. Furthermore, these models are accompanied by publicly available official implementations and pretrained model weights. To assess their performance against manipulation attacks, we utilize the pretrained model weights provided by the authors instead of training the models from scratch. For models with multiple versions, we select the one with the best performance reported.

Dataset. We used the Logical Access (LA) part of the ASVspoof 2019 dataset, which is commonly utilized by audio deepfake detection methods [13], [15], [34]–[37], including the baseline models. We followed baseline methods and trained our models exclusively on the training set of the ASVspoof 2019 LA to ensure a fair comparison. For evaluation, we used only the evaluation set of ASVspoof 2019 LA, excluding the other datasets such as ASVspoof 2021 LA and ASVspoof 2021 DF. This is due to the subpar performance of the pretrained baseline models on these datasets, making it challenging to demonstrate the impact of manipulation attacks.

Metrics. The evaluation metrics utilized in our study comprise the False Acceptance Rate (FAR), False Rejection Rate (FRR), F1 score, and Equal Error Rate (EER). FAR represents the proportion of deepfakes falsely classified as real audio, whereas FRR represents the opposite. Notably, FAR serves as a measure of the success rate of manipulation attacks and is therefore adopted as the primary metric for our evaluation. F1 score is computed as the harmonic mean of precision and recall. The EER corresponds to the point on the Detection Error Tradeoff (DET) curve where the FAR equals the FRR. Notably, the calculation of FAR, FRR, and F1 score necessitates a threshold, and for most evaluations, we adopt the threshold that yields the EER on the original data. Thus in our evaluation, FAR is also the EER on the original data.

Model hyperparameters. We take AASIST [15] model without the final fully connected layer as the encoder of our model to demonstrate the improvement of CLAD. The input audio is repeated or clipped to make the duration is fixed to 64600 samples. For pretraining stage, the model is trained with Adam optimizer using a learning rate of 0.0005, a weight decay of 0.0001, 150 epochs and a mini-batch size of 24. We also use the cosine annealing learning rate decay following the strategy of MoCo and AASIST. The queue size for contrastive learning is reduced to 6144, since large queue size will cause the queue to be filled with the same samples. The temperature τ and the momentum μ are set to 0.07 and 0.999, respectively. Concerning Length loss, we set the margin $margin$ to 4 and the weight w to 9. The weight of the length loss α is set to 2. The pretrain epochs is selected as 150 empirically. For downstream training stage, we train the model with Adam optimizer using a learning rate of 0.001, a weight decay of 0.0001, 10 epochs and a mini-batch size of 16.

Manipulation parameters To evaluate the impact of white noise, we varied the signal-to-noise ratio (SNR) with values of 15, 20, and 25dB. Regarding environmental noise, we added 7 types of noise from the ESC-50 dataset [38] at a constant SNR of 20dB. For volume control, we examined factors of 0.5 and 0.1. To investigate the effect of fading, we used a linear fade shape and varied the fade ratio with values of 0.1, 0.3, and 0.5. Additionally, we examined the influence of five fade shapes, each employing a consistent fade ratio of 0.5. To evaluate the impact of time stretching, we used an FFT length of 128 and varied the time stretching factor with values of 0.9, 0.95, 1.05, and 1.1. For resampling, we evaluated target resampling rates of 15,000, 15,500, 16,500, and 17,000 Hz, which correspond to -1,000, -500, +500, and +1,000 offsets to the original sampling rate of 16,000 Hz. We examined time shifting by considering shift lengths of 1,600, 16,000, and 32,000. To add echoes, we studied two parameters: delay and attenuation factor. We set the delay to 1,000 or 2,000 and the attenuation to 0.2 or 0.5. Additional information about the manipulation parameters is provided in the appendix.

TABLE I
THE FARS (%) AND F1 SCORES (%) OF DIFFERENT AUDIO DEEPFAKE DETECTION METHODS UNDER MANIPULATION ATTACKS.

Models	Types	None	Volume Control		White Noise			Environmental Noise[1]							Time Stretch			
			0.5	0.1	15dB	20dB	25dB	WD	FS	BR	CO	RA	CT	SN	1.1	1.05	0.95	0.9
RawNet2	FAR	4.60	2.78	**36.62**	**18.00**	10.09	7.44	4.02	2.95	3.74	4.30	8.68	7.65	3.37	0.28	0.44	0.33	0.16
	F1	81.08	86.90	**37.16**	**54.24**	67.42	73.38	82.86	86.34	83.74	81.99	70.46	72.88	84.94	96.43	95.78	96.24	96.95
AASIST	FAR	0.83	1.49	**9.12**	0.07	0.13	0.40	0.08	0.26	0.17	0.10	0.10	0.11	0.13	0.16	0.08	0.04	0.03
	F1	96.11	93.51	**71.25**	99.30	99.02	97.89	99.26	98.45	98.84	99.15	99.14	99.12	99.04	98.89	99.26	99.43	99.45
Res-TSSDNet	FAR	1.63	2.19	**10.11**	**51.28**	**39.23**	**29.80**	**36.68**	12.55	3.91	1.68	**40.37**	12.70	2.84	0.19	0.20	0.00	0.00
	F1	92.57	90.50	**68.74**	**30.56**	**36.49**	**43.03**	**38.05**	64.01	84.69	92.39	**35.84**	63.74	88.19	98.34	98.31	99.17	99.17
SAMO	FAR	1.09	3.11	**7.67**	0.22	0.77	1.40	0.38	0.29	0.58	0.12	0.58	0.32	0.46	0.00	0.01	0.00	0.00
	F1	94.94	87.56	**74.50**	98.51	96.22	93.74	97.83	98.19	97.01	98.94	96.98	98.09	97.48	99.44	99.43	99.45	99.45
CLAD	FAR	1.11	0.70	0.06	0.11	0.51	0.82	1.39	1.17	0.68	0.20	0.23	1.15	1.01	0.07	0.12	0.06	0.03
	F1	94.82	96.48	99.17	98.95	97.26	96.01	93.74	94.59	96.56	98.60	98.43	94.68	95.25	99.13	98.91	99.18	99.32

Models	Types	Add Echoes[2]			Time Shift			Fade[3]							Resample			
		1k/ .2	1k/ .5	2k/ .5	1.6k	16k	32k	.5/ L	.3/ L	.1/ L	.5/ E	.5/ Q	.5/ H	.5/ Lo	-1k	-0.5k	+0.5k	+1k
RawNet2	FAR	3.55	1.48	2.71	4.08	3.99	2.14	15.57	7.61	4.79	**27.09**	7.88	**30.42**	4.58	3.82	3.92	3.87	3.67
	F1	84.34	91.62	87.14	82.66	82.92	89.16	57.71	72.97	80.50	**44.30**	72.31	**41.52**	81.14	83.48	83.15	83.33	83.95
AASIST	FAR	0.23	0.07	0.10	0.60	0.75	0.54	9.63	3.60	1.05	**23.53**	3.83	**31.22**	1.23	0.59	0.60	0.97	1.25
	F1	98.59	99.29	99.16	97.05	96.45	97.31	70.13	86.08	95.24	**49.15**	85.32	**42.17**	94.51	97.07	97.03	95.55	94.43
Res-TSSDNet	FAR	2.18	1.29	1.61	2.38	2.25	2.15	2.80	1.68	1.17	4.91	1.83	**11.55**	1.47	0.00	0.00	1.78	2.08
	F1	90.53	93.88	92.66	89.83	90.29	90.64	88.34	92.39	94.35	81.62	91.82	**65.87**	93.17	99.18	99.18	92.01	90.89
SAMO	FAR	0.15	0.04	0.07	0.69	1.89	1.58	**17.65**	10.34	2.16	**25.98**	11.07	**28.24**	4.53	0.38	0.43	1.19	1.55
	F1	98.78	99.28	99.17	96.56	91.89	93.05	**56.17**	68.51	90.89	**46.60**	67.05	**44.53**	83.01	97.84	97.64	94.55	93.16
CLAD	FAR	0.21	0.03	0.07	0.85	0.93	0.89	0.64	0.95	1.06	0.50	0.85	0.80	0.93	0.65	0.81	1.37	1.63
	F1	98.55	99.32	99.15	95.88	95.57	95.74	96.73	95.50	95.05	97.32	95.88	96.07	95.57	96.68	96.03	93.81	92.83

[1] The abbreviations WD, FS, BR, CO, RA, CT, SN stand for wind, footsteps, breathing, coughing, rain, clock tick and sneezing respectively.
[2] 1k/ .2 means that the delay is 1,000 samples and the attenuation factor is 0.2, and the rest is the same.
[3] .5/ L means the ratio is set to 0.5, and using linear fade shape. L, E, Q, H, Lo denote linear, exponential, quarter sinusoidal, half sinusoidal, logarithmic fade shapes respectively.

B. Manipulation Attacks Results (RQ1)

To address the first research question, we conducted a large-scale experiment to investigate the impact of manipulations on the performance of the baseline models. Only the audio deepfakes are manipulated to bypass the detection. The results are presented in the first four rows of Tab. I.

It is evident that all baseline models exhibit excellent performance in the absence of manipulations, with FARs of 4.60%, 0.83%, 1.63% and 1.09% for RawNet2, AASIST, Res-TSSDNet and SAMO, respectively. However, simple volume control significantly increases the FAR, reaching 36.69% for RawNet2 when the control factor is set to 0.1, with similar increases observed for other baselines. Regarding noise injection, RawNet2 is vulnerable to white noise, while Res-TSSDNet experiences severe degradation from white noise, wind noise, and rain noise, as indicated by an FAR as high as 51.28%. Interestingly, AASIST and SAMO performs better, with a lower FAR even compared to the original data, indicating its ability to identify noisy audio as potential deepfakes.

When facing time stretch manipulation, all models consistently yield better results, likely due to the noticeable artifacts introduced by the FFT and inverse FFT processes in time stretching, which are frequently observed in audio deepfakes. For echoes and timeshift, none of the baselines are affected much. Fading with half sinusoidal fade shape is a strong manipulation that consistently achieving high bypass performance with FARs of 30.43%, 31.23%, and 11.55%, 28.24% for RawNet2, AASIST, Res-TSSDNet and SAMO, respectively. Resampling does not show significant influence, consistent with findings in [21] who tested a narrower range. Regarding F1 scores, we observe similar trends as FARs.

To examine how manipulation attacks influence the model's prediction score, we take three representative manipualtion attack settings as examples and analyze the score distribution for the entire dataset output by three baselines. For better visualization, we adopt the approach used by [13], which employs the second element of the final linear layer output as the prediction score, which reflects the softmax output. The distribution of prediction scores for the selected baseline models under various manipulations is presented in Fig. 5. Notably, higher scores indicate higher level of confidence in the authenticity of the sample. We observe that though the original score distribution without manipulation for different models are different, they all share a same trait that the distribution of deepfakes and real audio can be easily separated. However, after manipulation, we observe that the score distribution for the deepfakes shift towards distribution of real audio. Consequently, with the original threshold, we could expect a high FAR for the manipulated samples.

In order to enhance the effectiveness of attacks, a straightforward and intuitive strategy for adversaries is to combine multiple potent manipulations. We assess the performance of baseline models against a combination of manipulation attacks. We employ a representative set of manipulations for evaluation, including: volume control (VC) with a factor of 0.1, white noise (WN) with a SNR of 15 dB, wind environmental noise(EN), time stretch(TS) with a stretching factor of 0.9, fading(FD) with a half sine shape and fade ratio of 0.5, and resampling(RS) to 17,000 Hz. Tab. II presents the results. Each row represents the first manipulation applied, and each column represents the second. The diagonal elements represent the performance of the baseline models under single manipulation attacks. We observe that RawNet2 and Res-TSSDNet exhibit a significantly higher FAR under combined attacks. In contrast, for AASIST and SAMO, the combination of manipulations does not result in a higher FAR.

To sum up, our evaluation indicates that all baseline models are vulnerable to manipulation attacks. Although the baseline models exhibit varying degrees of vulnerability, volume control, noise injection, and fading generally achieve higher

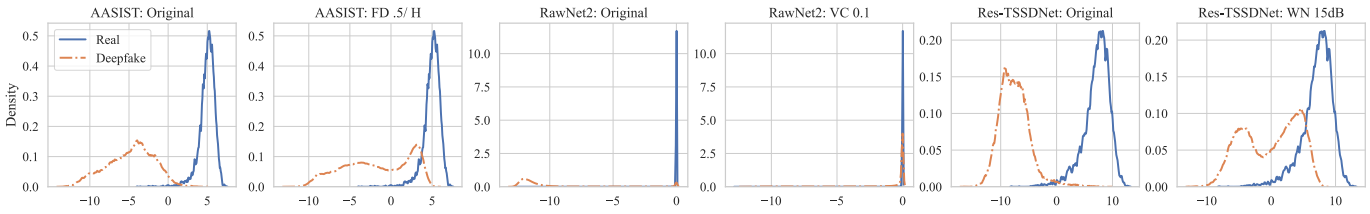


Fig. 5. The prediction score distribution of baseline model output for the whole dataset under various types of manipulations.

TABLE II
THE FARS (%) OF DIFFERENT AUDIO DEEPFAKE DETECTION METHODS UNDER COMBINATION OF MANIPULATION ATTACKS.

(a) RawNet2

	VC	WN	EN	TS	FD	RS
VC 0.1	36.69	86.47	59.9	7.87	33.57	34.07
WN 15dB	86.45	18.02	21.77	5.09	73.86	11.46
EN Wind	59.91	21.90	8.68	1.30	58.67	7.14
TS 0.9	7.89	6.53	1.67	0.16	1.31	0.09
FD .5/ H	33.57	71.7	56.02	1.01	30.43	27.35
RS +1k	34.07	20.03	9.14	0.09	26.02	3.67

(b) AASIST

	VC	WN	EN	TS	FD	RS
VC 0.1	9.15	12.64	10.61	0.45	20.71	9.83
WN 15dB	12.71	0.07	0.06	0.03	15.51	0.14
EN Wind	10.61	0.06	0.10	0.03	26.74	0.24
TS 0.9	0.45	0.03	0.03	0.03	0.31	0.03
FD .5/ H	20.71	0.96	3.28	0.59	31.23	30.57
RS +1k	9.83	0.16	0.23	0.07	32.14	1.25

(c) Res-TSSDNet

	VC	WN	EN	TS	FD	RS
VC 0.1	10.11	73.62	70.24	0.78	6.18	10.61
WN 15dB	73.63	51.28	54.53	0.00	49.17	52.57
EN Wind	70.24	54.43	40.37	0.00	43.00	42.12
TS 0.9	0.78	7.45	2.62	0.00	0.13	0.00
FD .5/ H	6.18	56.3	50.82	0.39	11.55	16.77
RS +1k	10.61	0.96	3.28	0.59	30.57	2.08

(d) SAMO

	VC	WN	EN	TS	FD	RS
VC 0.1	7.67	10.54	9.00	0.00	5.36	8.08
WN 15dB	10.46	0.22	0.15	0.00	21.48	0.27
EN Wind	9.00	0.15	0.38	0.00	26.48	0.79
TS 0.9	0.00	0.01	0.01	0.00	0.03	0.00
FD .5/ H	5.36	16.96	26.64	0.06	28.24	29.79
RS +1k	8.08	0.27	0.74	0.00	30.11	1.55

(e) CLAD

	VC	WN	EN	TS	FD	RS
VC 0.1	0.06	0.03	0.04	0.00	0.00	0.08
WN 15dB	0.02	0.12	0.05	0.01	0.33	0.38
EN Wind	0.00	0.00	0.23	0.00	0.19	0.31
TS 0.9	0.00	0.00	0.01	0.03	0.10	0.31
FD .5/ H	0.00	0.00	0.02	0.00	0.81	1.49
RS +1k	0.02	0.10	0.84	1.11	0.31	1.63

FARs. Furthermore, combining different manipulation techniques results in stronger attack performance for baselines like RawNet2 and Res-TSSDNet.

C. CLAD Performance (RQ2)

In this section, we evaluate CLAD's effectiveness against manipulation attacks and answer the second research question. Tab. I presents the results. We observe a slight increase in FAR from 0.83% to 1.11% for CLAD in the absence of manipulation. Notably, 0.83% represents the best result achieved by AASIST, with the authors reporting an average training result of 1.13% [15]. This increase in FAR is an acceptable trade-off, considering the overall robustness of CLAD. In the presence of white noise injection, the proposed CLAD model outperforms Res-TSSDNet, achieving an FAR of 0.12%, whereas Res-TSSDNet records a substantially higher FAR of 51.28%. Similarly, CLAD exhibits significant improvements over RawNet2 and AASIST when subjected to volume control and fading. Even for combinations of manipulations, CLAD maintains a lower FAR compared to using only a single manipulation, as depicted in Tab. II. In conclusion, CLAD demonstrates robustness against manipulation attacks by maintaining the highest FAR of 1.63% and the lowest F1 score of 92.82% among all tested manipulation attack scenarios.

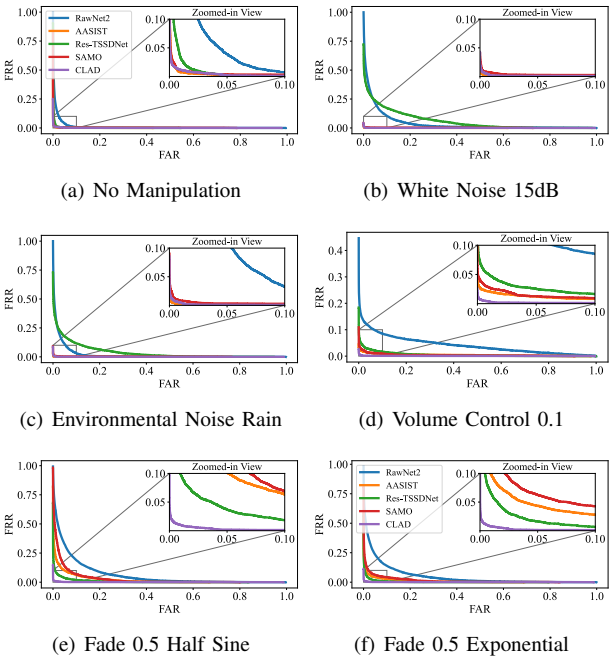
Additionally, we present the evaluation results using Detection Error Tradeoff (DET) curves, which offer a comprehensive visualization of the tradeoff between the False Acceptance Rate (FAR) and False Rejection Rate (FRR). Fig. 6 depicts the DET curves for different models across four manipulation attack settings. Curves positioned in the lower left quadrant indicate better detection performance. It can be observed that CLAD consistently performs excellently, while the baseline models exhibit varying degrees of performance degradation under certain manipulations.

It should be noted that CLAD can be integrated with existing end-to-end detection models by taking them as an encoder. In this study, we selected the AASIST architecture as the encoder for CLAD. To assess the impact of the encoder architecture, we conducted performance evaluations of CLAD using RawNet2 and Res-TSSDNet as alternative encoders. SAMO is based on the same architecture as AASIST, so we do not compare it here. Representative results are illustrated in Fig. 7, and full results can be found in Tab. III. The results demonstrate that the encoder architecture do influence the performance of CLAD. Specifically, the model employing an AASIST encoder consistently outperforms the alternative encoders across most experimental conditions. This observation aligns with the baseline model performance reported in Tab.I, suggesting that a better encoder architecture could enhance the performance of CLAD. Moreover, observing the results presented in Tab. III, it is noteworthy that the performance of CLAD with RawNet2 encoder still outperforms the performance of the original model under most manipulation attacks. We even trained a CLAD model with RawNet2 encoder with better performance on original data compared to the

TABLE III
THE FARS (%) OF MODEL WITH DIFFERENT ENCODER ARCHITECTURES UNDER ALL MANIPULATIONS

Encoder	None	Volume Control		White Noise			Environmental Noise							Time Stretch			
Architecture		0.5	0.1	15dB	20dB	25dB	WD	FS	BR	CO	RA	CT	SN	1.1	1.05	0.95	0.9
RawNet2	3.37	1.44	0.36	4.06	3.61	3.24	2.28	6.13	3.63	2.24	3.86	8.23	2.85	0.29	0.37	0.32	0.23
Res-TSSDNet	3.00	1.25	0.53	0.03	0.15	0.44	0.20	0.70	3.23	0.29	0.19	0.68	1.97	0.19	0.28	0.20	0.13
AASIST	1.11	0.70	0.06	0.12	0.52	0.78	1.39	1.17	0.68	0.20	0.23	1.15	1.01	0.08	0.12	0.06	0.03

Encoder	Add Echoes			Time Shift			Fade							Resample			
Architecture	1k/ .2	1k/ .5	2k/ .5	1.6k	16k	32k	.5/ L	.3/ L	.1/ L	.5/ E	.5/ Q	.5/ H	.5/ Lo	-1k	-0.5k	+0.5k	+1k
RawNet2	2.19	1.17	1.30	2.50	3.01	3.00	1.31	2.10	2.94	2.05	1.74	7.08	1.95	3.98	3.28	2.81	2.54
Res-TSSDNet	0.71	0.14	0.16	2.41	2.43	2.48	0.70	1.08	1.61	0.52	0.99	0.86	1.30	1.78	2.19	3.43	3.87
AASIST	0.21	0.03	0.07	0.85	0.93	0.89	0.64	0.95	1.06	0.50	0.85	0.81	0.93	0.65	0.81	1.38	1.63



(a) No Manipulation (b) White Noise 15dB

(c) Environmental Noise Rain (d) Volume Control 0.1

(e) Fade 0.5 Half Sine (f) Fade 0.5 Exponential

Fig. 6. DET curve of detection models under different manipulations.

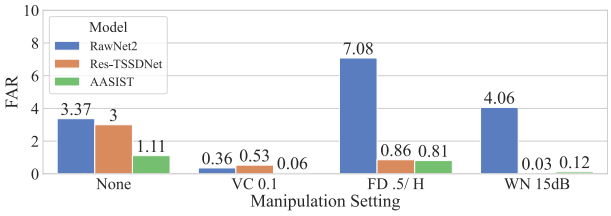


Fig. 7. The FARs (%) of models with differnet encoder architectures under different manipulation attacks

model released by the authors. Results here underscore the CLAD's effectiveness in improving the robustness of existing deepfake detection models. Fig. 8 presents an illustrative audio deepfake case alongside predictions generated by various detection methods under different manipulation settings for better comprehension.

D. Ablation Study (RQ3)

In order to systematically assess the key components of our approach, we conducted ablation experiments from two

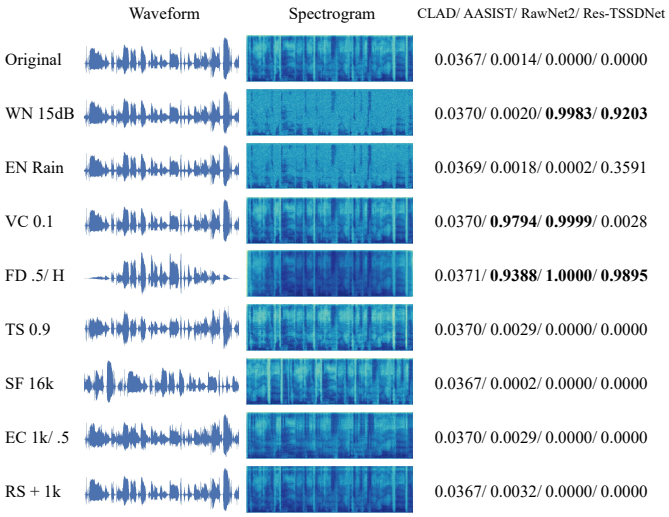


Fig. 8. Audio Deepfake examples under various manipulations, along with predicted probabilities of being real. Higher predictions indicate the model mistakenly identifies the audio as real.

perspectives and provide a comprehensive analysis in the following sections.

Model Variants. We consider four model variants to evaluate the components of CLAD, which are summarized in Tab. V. The ablation study results are presented in Tab. IV.

1) Vanilla. This is an AASIST model trained without contrastive learning and length loss. Compared with the baseline, it employs all manipulations as data augmentation.
2) CL. This represents our model trained without length loss.
3) LL. Refers to the Vanilla model trained with the inclusion of length loss as an additional loss function.
4) CLAD. Denotes our proposed method, which utilizes both contrastive learning and length loss.

Effectiveness of Contrastive Learning. Examining Tab. IV, it is evident that contrastive learning proves effective in enhancing the model's robustness against manipulation attacks. For models trained without contrastive learning, i.e., Vanilla and LL, we observe FAR values of 11.47% and 13.87% under fading and volume control, respectively. Even though these models are trained with all the manipulations, conventional supervised learning struggles to handle such diverse manipulations effectively. In contrast, CL and CLAD yield more consistent results across all manipulations. It is worth noting that the FAR of LL for the original dataset

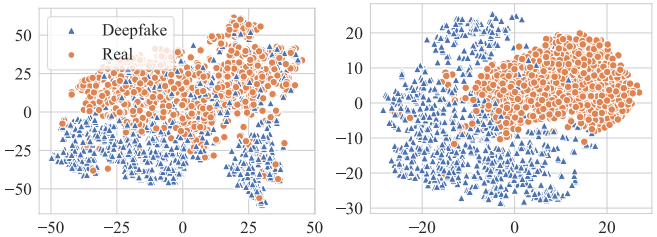
TABLE IV
THE FARS (%) OF MODEL VARIANTS IN ABLATION STUDY UNDER ALL MANIPULATIONS.

Models	None	Volume Control		White Noise			Environmental Noise							Time Stretch			
		0.5	0.1	15dB	20dB	25dB	WD	FS	BR	CO	RA	CT	SN	1.1	1.05	0.95	0.9
Vanilla	3.02	4.80	0.19	0.12	0.71	1.67	3.19	4.12	2.87	0.99	0.69	4.96	2.98	0.39	0.53	0.31	0.06
CL	4.36	1.09	0.03	2.52	4.20	4.61	4.20	4.23	4.10	1.50	3.01	5.89	3.43	0.79	0.90	0.88	0.67
LL	2.00	13.87	2.24	6.63	4.17	2.72	4.26	5.84	4.17	13.70	4.76	7.29	4.94	9.64	7.04	7.34	9.72
CLAD	1.11	0.70	0.06	0.12	0.52	0.78	1.39	1.17	0.68	0.20	0.23	1.15	1.01	0.08	0.12	0.06	0.03

Models	Add Echoes			Time Shift			Fade							Resample			
	1k/ .2	1k/ .5	2k/ .5	1.6k	16k	32k	.5/ L	.3/ L	.1/ L	.5/ E	.5/ Q	.5/ H	.5/ Lo	-1k	-0.5k	+0.5k	+1k
Vanilla	0.55	0.04	0.16	1.90	2.06	1.93	6.14	5.03	3.47	6.87	5.46	11.47	4.79	2.32	2.63	3.20	4.11
CL	1.85	0.76	1.02	3.73	3.79	3.87	1.82	3.33	4.08	1.25	2.79	1.95	3.19	3.44	3.97	4.94	5.88
LL	3.73	8.12	9.46	2.79	2.92	2.90	2.52	2.15	2.04	3.33	2.26	4.43	2.09	2.34	2.26	2.22	2.54
CLAD	0.21	0.03	0.07	0.85	0.93	0.89	0.64	0.95	1.06	0.50	0.85	0.81	0.93	0.65	0.81	1.38	1.63

TABLE V
THE MODEL VARIANTS IN ABLATION STUDY

Component	Vanilla	CL	LL	CLAD
Contrastive Learning	✗	✓	✗	✓
Length Loss	✗	✗	✓	✓



(a) w/o length loss (b) w/ length loss

Fig. 9. Visualization of the features extracted using the encoders trained without and with length loss.

stands at 4.36%, which is less favorable compared to other models. This discrepancy is attributed to the unsupervised nature of contrastive learning, which results in the model being insufficiently trained with label information.

Effectiveness of Length Loss. Length loss encourages the model to learn the clustering of real audio, thus making downstream training easier. As demonstrated in Tab. IV, models trained with length loss outperform those without it on the original data. LL achieves an FAR of 2.00%, while Vanilla achieves 3.02%. Notably, CLAD shows a substantial improvement compared to CL, with CLAD achieving a FAR of 1.11% compared to CL's 4.36%. Similar improvement can also be observed under various manipulations. Such enhancements can be attributed to the design of the length loss, which utilizes length of feature vectors and complements contrastive learning. To provide further insight into the impact of length loss, we visualize the feature vectors extracted by the encoder, both trained with and without length loss, in Fig. 9. The visualization illustrates that the encoder trained with length loss produces feature vectors that are more closely aggregated. This, in turn, enhances detection performance.

TABLE VI
THE FARS (%) OF MODELS TRAINED WITH PARTIAL MANIPULATIONS UNDER DIFFERENT MANIPULATION ATTACKS INCLUDING UNKNOWN MANIPULATION.

	VC	WN	EN	TS	EC	SF	FD	RS
None	1.49	2.61	1.96	2.03	1.76	1.26	2.24	2.11
VC 0.1	7.39	2.77	2.05	1.70	2.90	1.69	0.49	4.25
WN 15dB	0.50	1.12	0.16	0.52	0.40	0.19	0.37	0.36
EN Rain	0.80	1.65	0.59	0.93	0.83	0.58	0.84	0.90
TS 0.9	0.25	0.52	0.28	0.03	0.34	0.03	0.21	0.22
EC 1k/ .2	0.67	1.19	0.73	0.67	0.28	0.30	0.96	0.50
SF 16k	1.21	2.20	1.55	1.74	1.45	0.87	1.97	1.81
FD .5/ H	2.81	3.22	2.30	3.32	3.60	2.34	6.19	2.00
RS +1k	1.77	3.52	2.75	2.31	2.32	1.72	2.90	3.04

E. Unknown Manipulation Study (RQ4)

In this section, we assess the detection capability of CLAD for unknown manipulation attack methods, considering that the manipulation methods discussed in our paper may not encompass all methods used by attackers. Of the eight manipulation methods examined in our study, we trained eight CLAD models, each with one manipulation method removed. Subsequently, we assessed the performance of these models when confronted with unknown manipulation attacks. The representative results are presented in Tab. VI, each column represents a model trained without a specific manipulation.

We observed that volume control and fading are particularly strong attacks, as models trained without these manipulations performed not well, resulting in a FAR of 7.19% and 6.19%, respectively. However, for other manipulations, models trained without them still achieved good performance under unknown manipulation attacks. For instance, the model trained without time stretch achieved a low FAR of 0.03% under time stretch manipulation. Similar results were observed for time shift and echoes adding. Moreover, compared to baseline models, CLAD demonstrated improved performance against unknown manipulations in most scenarios. For example, although both trained without fading manipulation, our design enabled CLAD to reduce its FAR from 31.22% to 6.19% compared with baseline AASIST model. Therefore, we conclude that while strong manipulations like volume control and fading may lead to suboptimal results, CLAD is expected to outperform baseline models and achieve good performance against most unknown manipulation attacks.

F. Key Findings

- **Manipulation attacks pose a severe threat to deepfake detection models.** We found that all the baseline models we evaluated are vulnerable to particular manipulations. Among manipulations, volume control and fading demonstrate the most favorable attack performance. While volume control can be mitigated through straightforward techniques like input normalization, fading manipulations pose a significant challenge to current detection methods. Fading with a half sine fade shape demonstrates a notably high 25.36% average FAR across the four baseline models.
- **CLAD enhances model resilience against manipulations.** We successfully trained a robust audio deepfake detection model with AASIST encoder architecture, achieving a low 1.63% FAR under all manipulations. Furthermore, our experiments confirm that CLAD can serve as a plug-and-play module to enhance the robustness of existing deepfake detection models. Notably, CLAD trained with the RawNet2 encoder outperforms the model released by the authors, even in unmanipulated data.
- **Contrastive Learning and Length Loss are essential components of CLAD.** We found that contrastive learning trained model perform more consisitently under different manipulation attacks, indicating its improved robustness. Length loss is also an essential component of CLAD, complementing contrastive learning and further improving its performance. Moreover, our evaluation results suggest that intuitive data augmentation is not the optimal solution for countering manipulation attacks.
- **CLAD shows promise for superior performance against unknown manipulation attacks.** Our study reveals that CLAD demonstrates promising performance in the presence of unknown manipulation attacks. Even against strong manipulations like volume control and fading, it outperforms baseline models.

VI. Related Work

A. Contrastive Learning

Contrastive learning is a popular form of self-supervised learning [39], [40] that encourages augmentations (views) of the same input to have more similar representations than augmentations of different inputs. Earlier studies [23], [26] illustrates the effectiveness of contrast learning by showing that contrastive learning can even yield better results than supervised learning in some cases. Although contrastive learning has been a great success in the field of computer vision [41]–[43], it has not received enough attention in the audio domain. Saeed et al. [24] proposed COLA, a self-supervised pre-training approach for learning a general-purpose representation of audio. However, COLA does not discuss the downstream tasks of audio deepfake detection. Guan et al. [25] introduced contrastive learning into the anomalous sound detection and showed its effectiveness. In conclusion, existing work on contrast learning is not sufficient for the audio domain, and as far as we know, our work is the first to apply contrast learning to the field of audio deepfake detection.

B. Audio Deepfake Detection

The rise of audio deepfakes has posed significant security threats, prompting a surge of interest in developing robust detection methods [30], [44], [45]. Earlier approaches [11], [12], [46] were mainly based on handcrafted features and machine learning-based classifiers. In 2021, Tak et al. [13] proposed a milestone model which is based on an end to end speaker verification model, RawNet2 [47]. Remarkably, RawNet2 demonstrated state-of-the-art (SOTA) performance without the need for hand-crafted features. Its pre-trained models quickly gained popularity within the field. After that, Jung et al. [15] proposed a heterogeneous stacking graph attention layer that models artefacts spanning heterogeneous temporal and spectral intervals, and achieved SOTA performance on the ASVspoof 2019 dataset. Blue et al. [48] leverage fluid dynamics to estimate the human vocal tract arrangement during speech production and use it to identify deepfakes. This approach, however, needs accurate phoneme timestamps, which are seldom accessible in practical scenarios.

Recently, instead of solely focusing on benchmark datasets performance, researchers have begun considering broader aspects of audio deepfake detection. Recognizing the limitations of existing models on non-English data, Ba et al. [31] proposed domain adaptation strategies for cross-lingual detection. Zhang et al. [49], [50] investigated continual learning in audio deepfake detection, which is an effective way to help detection models adapt to new attack types. Zhang et al. [33] and Singh et al. [51] investigated the challenges in detecting audio deepfakes in compressed form.

However, most of these approaches do not consider the scenarios where the attacker manipulate the speech. A notable exception is DeepSonar [21], which evaluates the robustness of the model and performs a small-scale experiment of manipulation attacks. In this work, we conduct a larger scale experiment on ASVSpoof 2019 evaluation set (containing 71,237 samples) with more types of manipulation attacks.

VII. Discussion

Intuitive Defense. An intuitive defense against manipulation attacks is denoising, as noise injection is one of manipulations. However, our preliminary experiment with Wiener filtering does not yield satisfactory outcomes. While modern speech enhancement methods have potentials, we are concerned that enhancements artefact might trigger false alarms in the detection model. We leave further investigation into this matter to future research.

Diverse Datasets. In this study, we opt to evaluate the baselines using the parameters provided by the authors to make the results more convincing. However, this choice limited our dataset to the ASVspoof 2019 dataset, as the pretrained models did not perform well on other publicly available datasets. With the advancement of the field, we anticipate that future models will exhibit improved generalization and adaptable to a wider range of datasets, enabling evaluations on a more diverse dataset in the future.

VIII. Conclusion

In this paper, we expose the threat of manipulation attacks against audio deepfake detection systems. It does not require prior knowledge of the system, making it simple for deepfake creators to degrade the detection system performance. To address this issue, we conduct a large-scale evaluation of widely adopted models on the ASVspoof dataset under manipulation attacks and find that existing SOTA models experience severe performance degradation when faced with manipulation attacks. To mitigate the impact of manipulation attacks, we propose contrastive learning-based detection approach and design length loss to train a model to produce robust representations against manipulations. Experimental results show that CLAD achieves significantly better performance than the widely adopted models under manipulation attacks.

References

[1] Y. Wang, R. J. Skerry-Ryan *et al.*, "Tacotron: Towards End-to-End Speech Synthesis," in *INTERSPEECH 2017*, F. Lacerda, Ed. ISCA, 2017, pp. 4006–4010.

[2] R. Huang, Z. Zhao *et al.*, "ProDiff: Progressive Fast Diffusion Model for High-Quality Text-to-Speech," in *Proceedings of the 30th ACM International Conference on Multimedia (MM)*. Lisboa Portugal: ACM, October 2022, pp. 2595–2605.

[3] Y. A. Li, C. Han *et al.*, "StyleTTS 2: Towards Human-Level Text-to-Speech through Style Diffusion and Adversarial Training with Large Speech Language Models," in *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems (NeurIPS) 2023*, 2023.

[4] Y. A. Li, C. Han, and N. Mesgarani, "Styletts-VC: One-Shot Voice Conversion by Knowledge Transfer From Style-Based TTS Models," in *2022 IEEE Spoken Language Technology Workshop (SLT)*, 2023, pp. 920–927.

[5] H. Tang, X. Zhang *et al.*, "Avqvc: One-Shot Voice Conversion By Vector Quantization With Applying Contrastive Learning," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, May 2022, pp. 4613–4617.

[6] J. Li, W. Tu, and L. Xiao, "Freevc: Towards High-Quality Text-Free One-Shot Voice Conversion," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023.

[7] S. Khatsenkova, "Audio deepfake scams: Criminals are using ai to sound like family and people are falling for it," 2023. [Online]. Available: https://www.euronews.com/embed/2231732

[8] E. Wenger, M. Bronckers *et al.*, ""Hello, It's Me": Deep Learning-Based Speech Synthesis Attacks in the Real World," in *Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, New York, NY, USA, 2021, pp. 235–251.

[9] Z. Khanjani, G. Watson, and V. P. Janeja, "Audio deepfakes: A survey," *Frontiers Big Data*, vol. 5, 2022.

[10] E. A. AlBadawy, S. Lyu, and H. Farid, "Detecting AI-Synthesized Speech Using Bispectral Analysis," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, June 2019.

[11] G. Lavrentyeva, S. Novoselov *et al.*, "STC Antispoofing Systems for the ASVspoof2019 Challenge," in *INTERSPEECH 2019*. ISCA, September 2019, pp. 1033–1037.

[12] M. Todisco, H. Delgado, and N. Evans, "Constant Q cepstral coefficients: A spoofing countermeasure for automatic speaker verification," *Computer Speech & Language*, vol. 45, pp. 516–535, Sep. 2017.

[13] H. Tak, J. Patino *et al.*, "End-to-End anti-spoofing with RawNet2," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2021, pp. 6369–6373.

[14] G. Hua, A. B. J. Teoh, and H. Zhang, "Towards End-to-End Synthetic Speech Detection," *IEEE Signal Processing Letters*, vol. 28, pp. 1265–1269, 2021.

[15] J.-w. Jung, H.-S. Heo *et al.*, "AASIST: Audio Anti-Spoofing Using Integrated Spectro-Temporal Graph Attention Networks," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022, pp. 6367–6371.

[16] X. Wang, J. Yamagishi *et al.*, "ASVspoof 2019: A large-scale public database of synthesized, converted and replayed speech," *Computer Speech & Language*, vol. 64, p. 101114, November 2020.

[17] J. Yamagishi, X. Wang *et al.*, "ASVspoof 2021: accelerating progress in spoofed and deepfake speech detection," in *Proc. ASVspoof Challenge workshop*, 2021, pp. 47–54.

[18] Y. Zhang, Z. Jiang *et al.*, "Black-Box Attacks on Spoofing Countermeasures Using Transferability of Adversarial Examples," in *INTERSPEECH 2020*. ISCA, October 2020, pp. 4238–4242.

[19] S. Liu, H. Wu *et al.*, "Adversarial Attacks on Spoofing Countermeasures of Automatic Speaker Verification," in *IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*. IEEE, 2019, pp. 312–319.

[20] M. Panariello, W. Ge *et al.*, "Malafide: a novel adversarial convolutive noise attack against deepfake and spoofing detection systems," in *INTERSPEECH 2023*. ISCA, Aug. 2023, pp. 2868–2872.

[21] R. Wang, F. Juefei-Xu *et al.*, "DeepSonar: Towards Effective and Robust Detection of AI-Synthesized Fake Voices," in *Proceedings of the 28th ACM International Conference on Multimedia (MM)*. Seattle WA USA: ACM, October 2020, pp. 1207–1216.

[22] H. Hojjati and N. Armanfard, "Self-Supervised Acoustic Anomaly Detection Via Contrastive Learning," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, May 2022, pp. 3253–3257.

[23] K. He, H. Fan *et al.*, "Momentum Contrast for Unsupervised Visual Representation Learning," in *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. Seattle, WA, USA: IEEE, June 2020, pp. 9726–9735.

[24] A. Saeed, D. Grangier, and N. Zeghidour, "Contrastive Learning of General-Purpose Audio Representations," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. Toronto, ON, Canada: IEEE, June 2021, pp. 3875–3879.

[25] J. Guan, F. Xiao *et al.*, "Anomalous Sound Detection Using Audio Representation with Machine ID Based Contrastive Learning Pretraining," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023.

[26] T. Chen, S. Kornblith *et al.*, "A Simple Framework for Contrastive Learning of Visual Representations," in *Proceedings of the 37th International Conference on Machine Learning (ICML)*. PMLR, November 2020, pp. 1597–1607.

[27] Y. Zhang, F. Jiang, and Z. Duan, "One-Class Learning Towards Synthetic Voice Spoofing Detection," *IEEE Signal Process. Lett.*, vol. 28, pp. 937–941, 2021.

[28] S. Ding, Y. Zhang, and Z. Duan, "SAMO: Speaker Attractor Multi-Center One-Class Learning For Voice Anti-Spoofing," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, Jun. 2023, pp. 1–5.

[29] J.-w. Jung, H. Tak *et al.*, "SASV 2022: The First Spoofing-Aware Speaker Verification Challenge," in *INTERSPEECH 2022*. ISCA, Sep. 2022, pp. 2893–2897.

[30] J. Yi, R. Fu *et al.*, "ADD 2022: the first Audio Deep Synthesis Detection Challenge," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 9216–9220.

[31] Z. Ba, Q. Wen *et al.*, "Transferring Audio Deepfake Detection Capability across Languages," in *Proceedings of the ACM Web Conference (WWW) 2023*. Austin TX USA: ACM, Apr. 2023, pp. 2033–2044.

[32] N. M. Müller, P. Czempin *et al.*, "Does Audio Deepfake Detection Generalize?" in *INTERSPEECH 2022*. ISCA, 2022, pp. 2783–2787.

[33] J. Zhang, X. Yi, and X. Zhao, "A Compressed Synthetic Speech Detection Method with Compression Feature Embedding," in *INTERSPEECH 2023*. ISCA, Aug. 2023, pp. 5376–5380.

[34] Y. Wen, Z. Lei *et al.*, "Multi-Path GMM-MobileNet Based on Attack Algorithms and Codecs for Synthetic Speech and Deepfake Detection," in *INTERSPEECH 2022*. ISCA, Sep. 2022, pp. 4795–4799.

[35] X. Wang and J. Yamagishi, "Investigating Self-Supervised Front Ends for Speech Spoofing Countermeasures," in *The Speaker and Language Recognition Workshop (Odyssey 2022)*. ISCA, June 2022, pp. 100–106.

[36] E. Conti, D. Salvi *et al.*, "Deepfake Speech Detection Through Emotion Recognition: A Semantic Approach," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022, pp. 8962–8966.

[37] T.-P. Doan, L. Nguyen-Vu *et al.*, "BTS-E: Audio Deepfake Detection Using Breathing-Talking-Silence Encoder," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, Jun. 2023, pp. 1–5.

[38] K. J. Piczak, "ESC: Dataset for Environmental Sound Classification," in *Proceedings of the 23rd Annual ACM Conference on Multimedia (MM)*. ACM Press, 2015, pp. 1015–1018.

[39] X. Liu, F. Zhang *et al.*, "Self-Supervised Learning: Generative or Contrastive," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 1, pp. 857–876, Jan. 2023.

[40] L. Jing and Y. Tian, "Self-Supervised Visual Feature Learning With Deep Neural Networks: A Survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 43, no. 11, pp. 4037–4058, Nov. 2021.

[41] M. Kang and J. Park, "ContraGAN: Contrastive Learning for Conditional Image Generation," in *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*, H. Larochelle, M. Ranzato *et al.*, Eds., 2020.

[42] S. Park, J. Lee *et al.*, "Fair Contrastive Learning for Facial Attribute Classification," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2022, New Orleans, LA, USA, June 18-24, 2022*. IEEE, 2022, pp. 10 379–10 388.

[43] P. Wang, K. Han *et al.*, "Contrastive Learning Based Hybrid Networks for Long-Tailed Image Classification," in *IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2021, virtual, June 19-25, 2021*. Computer Vision Foundation / IEEE, 2021, pp. 943–952.

[44] J. Yi, J. Tao *et al.*, "ADD 2023: the Second Audio Deepfake Detection Challenge," in *Proceedings of the Workshop on Deepfake Audio Detection and Analysis co-located with 32th International Joint Conference on Artificial Intelligence (IJCAI 2023), Macao, China, August 19, 2023*, vol. 3597, 2023, pp. 125–130.

[45] J. Frank and L. Schönherr, "WaveFake: A Data Set to Facilitate Audio Deepfake Detection," in *Proceedings of the Neural Information Processing Systems (NeurIPS) Track on Datasets and Benchmarks 1*, 2021.

[46] M. E. Ahmed, I.-Y. Kwak *et al.*, "Void: A fast and light voice liveness detection system," in *29th USENIX Security Symposium (USENIX Security 20)*, 2020, pp. 2685–2702.

[47] J.-w. Jung, S.-b. Kim *et al.*, "Improved RawNet with Feature Map Scaling for Text-Independent Speaker Verification Using Raw Waveforms," in *INTERSPEECH 2020*. ISCA, Oct. 2020, pp. 1496–1500.

[48] L. Blue, K. Warren *et al.*, "Who Are You (I Really Wanna Know)? Detecting Audio DeepFakes Through Vocal Tract Reconstruction," in *31st USENIX Security Symposium (USENIX Security 22)*, 2022.

[49] X. Zhang, J. Yi *et al.*, "Do You Remember? Overcoming Catastrophic Forgetting for Fake Audio Detection," in *Proceedings of the 40th International Conference on Machine Learning (ICML)*. PMLR, Jul. 2023, pp. 41 819–41 831, iSSN: 2640-3498.

[50] ——, "What to Remember: Self-Adaptive Continual Learning for Audio Deepfake Detection," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 17, pp. 19 569–19 577, Mar. 2024.

[51] A. K. Singh Yadav, Z. Xiang *et al.*, "ASSD: Synthetic Speech Detection in the AAC Compressed Domain," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023.



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APPENDIX

MANIPULATION DETAILS

In the following section, we provide a description of the manipulation process and define the parameters used. For more detailed information, readers are encouraged to refer to our implementation(https://github.com/CLAD23/CLAD).

In the case of noise injection, we generate noise and adjust its amplitude to achieve a desired signal-to-noise ratio (SNR). A higher SNR corresponds to a lower level of added noise. In the context of environmental noise injection, we empirically identified the most effective environmental noise source from the ESC-50 dataset.

Volume control involves multiplying the amplitude of the audio waveform by a specified factor. The factor dictates the extent and direction of volume adjustment, whether it entails an increase or decrease, and the magnitude of the change.

Fading involves applying a specific fade shape to attenuate portions of audio at both the beginning and end. The fade shape defines a mask which is applied as a multiplier to the audio's amplitude. The fade ratio determines the extent of audio fading. For instance, a ratio of 0.5 implies that half of the audio at the beginning and half at the end will be faded. The maximum selectable ratio is also 0.5. Detailed information regarding the fade shape definition can be found in the PyTorch implementation.

Time stretching is performed using the official PyTorch implementation. This process alters the audio signal's duration while preserving its pitch. The FFT length is utilized in the FFT calculations involved. The time stretching factor specifies the ratio between the length of the stretched audio and the original audio. A factor less than 1 speeds up the audio.

Resampling is conducted using the official PyTorch implementation. This operation adjusts the audio signal's sample rate to a specified target rate. The target resampling rates defines the sample rate after resampling. When resampled audio is played at the original sample rate, both pitch and duration are affected.

Time shifting involves shifting the audio signal in the time domain by moving all samples forward or backward. The shift length specifies both the direction of audio signal movement and the number of samples shifted.

Echoes adding creates a delayed and attenuated copy of the original audio and adds them together. The delay parameter determines the echo's delay in samples, and the attenuation factor represents the number by which the echo's amplitude is multiplied.

All manipulations were implemented to process the raw waveform and output the raw waveform.

Exploiting Style Latent Flows for Generalizing Deepfake Video Detection

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Abstract

This paper presents a new approach for the detection of fake videos, based on the analysis of style latent vectors and their abnormal behavior in temporal changes in the generated videos. We discovered that the generated facial videos suffer from the temporal distinctiveness in the temporal changes of style latent vectors, which are inevitable during the generation of temporally stable videos with various facial expressions and geometric transformations. Our framework utilizes the StyleGRU module, trained by contrastive learning, to represent the dynamic properties of style latent vectors. Additionally, we introduce a style attention module that integrates StyleGRU-generated features with content-based features, enabling the detection of visual and temporal artifacts. We demonstrate our approach across various benchmark scenarios in deepfake detection, showing its superiority in cross-dataset and cross-manipulation scenarios. Through further analysis, we also validate the importance of using temporal changes of style latent vectors to improve the generality of deepfake video detection.

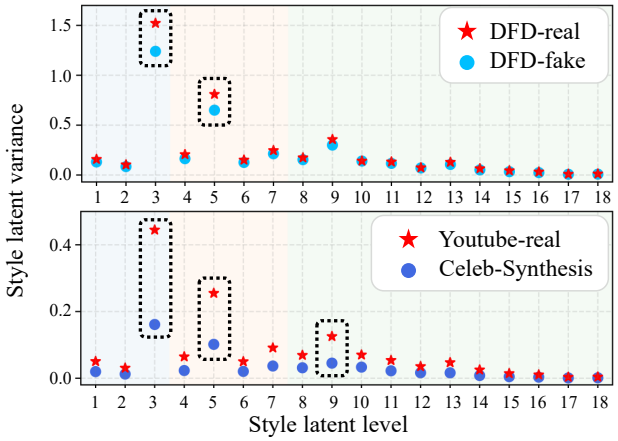


Figure 1. **Variance of style flow for each style latent level.** The x-axis shows the level of style latent vectors for fine style representations. We noticed that the level-wise differences vary across deepfake domains, but the variance of style latent vectors is particularly lower in certain levels of the style latent vectors for fake videos than in real videos. This happens due to the temporal smoothness of the style latent vectors to create temporally stable deepfake videos, and our results demonstrate that deepfake videos have a distinct variance in style flow compared to real videos.

1. Introduction

Recent generative algorithms are capable of producing high-quality videos; while this advancement has led to social concerns, as it becomes increasingly challenging to distinguish between generated videos and authentic ones. Generative models have the potential to expedite industries such as entertainment, gaming, fashion, design, and education. However, their misuse can have adverse effects on society. The high quality of the videos intensifies the potential for inappropriate utilization of the technique. To resolve the issue, researchers are actively engaged in developing the deepfake video detection algorithms [19, 29, 40].

Early deepfake detection research addressed spatial artifacts such as unnatural aspects [30] and frequency-level checkerboard of the generative model [33] in the generation of a single frame. While spatial artifact-based detection methods have shown reasonable performance for single images, they failed in accounting for temporal artifacts in deepfake videos that consist of multiple frames. To address the limitation, recent studies [15, 41] integrate temporal cues such as flickering [14, 57] and discontinuity [16] to enhance the accuracy of deepfake video detection task.

Nevertheless, existing methods [28, 51] that exploit visual and temporal cues of generated videos encountered performance degradation when attempting to identify videos produced by recent deepfake generation algorithms, which effectively suppress visual and temporal artifacts. Recent observations [36, 39] have presented a decline in the effectiveness of visual artifact-based algorithms against newly developed high-quality generation techniques [11, 37]. Temporal artifact-based algorithms [16] also experi-

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enced performance degradation due to the advanced quality of video generation methods [41, 57].

In this paper, we propose to focus on the suppressed variance in the temporal flow of style latent features which are extracted from generated videos, rather than conventional visual and temporal artifacts. The style latent feature encodes the facial appearance and facial movements, such as expressions and geometric transformations, to control the synthesis of facial images. Fig. 1 shows the clear differences in the flow of style latent vectors between the generated and synthetic videos. We noticed that the suppressed variance of the style latent vector occurs due to the temporal smoothness of facial appearance and movements in the generated videos. We believe that the flow of style latent vectors has the potential to be a distinctive feature for generalizing deepfake video detection across various generative models.

In order to encode the dynamics of style latent vectors, we introduce the StyleGRU module in our video deepfake detection framework. This module can effectively capture the temporal variations within style latent vectors, consequently enabling the extraction of robust style-based temporal features. To increase the representation power of style-based temporal features, we further involve the supervised contrastive learning [23] technique in StyleGRU module. We develop the style attention module which effectively integrates the style-based temporal features with the content features that encode the visual and temporal artifacts.

We perform various experiments to evaluate the proposed algorithm for multiple scenarios such as cross-dataset and cross-manipulation scenarios. From the experimental results, our method notably surpasses the existing algorithms, achieving a state-of-the-art performance. Ablation studies are also conducted to show the module-wise effectiveness, and the meaningful analysis experiments confirm that the variations in style latent vector contribute effectively to the robust detection to the various deepfake generation methods. The contributions of our work are summarized as follows.

- We propose a novel video deepfake detection framework that is based on the unnatural variation of the style latent vectors.
- We introduce a StyleGRU module that can encode the variations of style latent vectors with the contrastive learning technique.
- We propose a style attention module that effectively integrates the features from StyleGRU with the content features capturing conventional visual and temporal artifacts.
- Our approach demonstrates state-of-the-art performance in various deepfake detection scenarios, including cross-dataset and cross-manipulation settings, which confirms the effectiveness of facial attribute changes for deepfake video detection.

2. Related work

Deepfake detection. In deepfake detection research, there are two main types: image-level and video-level detectors. The image-level detectors detect fake images by recognizing spatial artifacts of a single frame. A straightforward approach to detect spatial artifacts involves using Xception model [40], a Convolutional Neural Network (CNN) architecture along with MAT [56] that utilizes an attentional mechanism. Face X-ray [27] proposed a method that uses boundaries between forged faces and backgrounds to capture spatial inconsistency. Several methods utilized the frequency-level features to emphasize the artifacts from the image generators [14]. FrePGAN [18] utilized the frequency map transformed from the original RGB image, and LRL [6] integrated RGB images with frequency-aware cues to learn inconsistencies. Recently, algorithms utilize methods including blending artifacts [2, 5, 12, 43] or separating elements relevant to detection from irrelevant ones [12, 55] during their training. Image-level detectors demonstrate high generalization performance. However, image-based methods have the limitation of not utilizing the temporal cues that inevitably occur during the generation of deepfake videos.

In contrast to image-level detectors, video-level detectors take advantage of temporal information by using multiple frames to detect deepfake videos. CNN-GRU [42] captured temporal information from the features extracted by CNNs, and LipForensics [16] utilized the information near the mouth region by employing a network pre-trained on a LipReading dataset. Recently, FTCN [57] directly extracted temporal information using 3D CNNs with a spatial kernel size of 1. AltFreeze [52] showcases strong generalization capabilities through its unique approach of independently training spatial and temporal information. However, the existing methods have focused on low-level features like pixels, so they ignore the temporal changes of high-level contexts such as facial attributes. To improve detection capabilities, we propose a novel approach that considers temporal features at both low-level and high-level, thereby enhancing the overall context and advancing the state-of-the-art in deepfake detection.

Style Latent Vectors. Well-trained Generative Adversarial Network (GAN) models [20–22] successfully generate disentangled latent space. This advancement permits the straightforward manipulation of style latent features to produce desired facial images. The GAN inversion, initially introduced by [59], targets the identification of a latent code within the latent space, which makes it possible to generate an image closely resembling a target one. Furthermore, various encoders [1, 38, 48, 58] have been studied to properly utilize the prior knowledge of StyleGAN latent space. Moreover, GAN inversion methods decoding the style latent features of StyleGAN [21] are utilized for generating the

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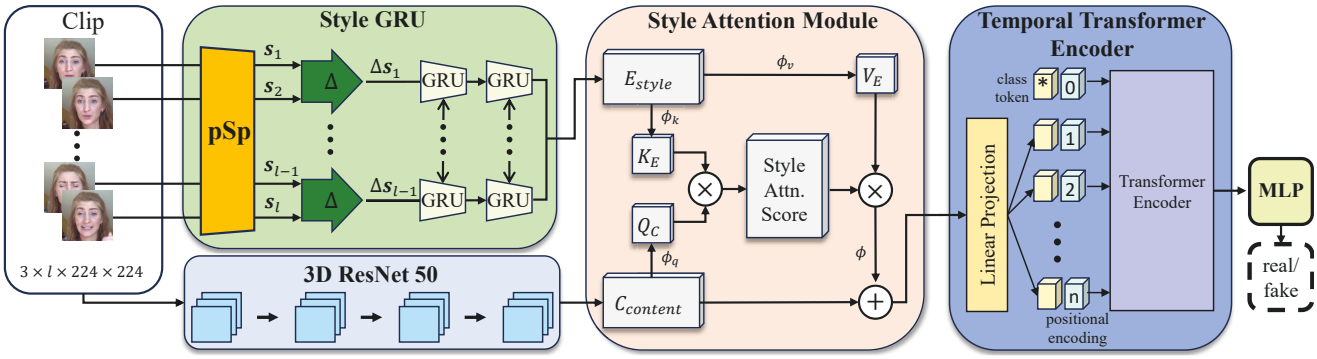


Figure 2. **Schematic diagram of our entire framework.** For the video deepfake detection, we first extract the style latent vectors from pSp for each individual frame, and then encode their variations using the StyleGRU module into the style-based temporal feature E_{style}. In parallel, the content feature C_{content} is extracted via the 3D ResNet-50 architecture from the video clip. Style Attention Module (SAM) integrates the style-based temporal feature E_{style} with the content feature C_{content} via the attention mechanism. Finally, Temporal Transformer Encoder (TTE) is applied to map the representation into the binary class of real and fake labels.

high-quality face-swap videos. MegaFS [60] also inverted both source and target images into the finely-tuned StyleGAN latent space for the purpose of face swap.

Face swapping through GAN inversion offers the advantage of separating facial identity in the latent space and swapping latent features, thereby preserving fidelity in models like StyleGAN [20]. Furthermore, it is employed for various Deepfake generation techniques, including modifying facial attributes, extending beyond just altering facial identity. These approaches surpassed the existing method, such as StyleRig [45], which involved direct modifications to the style latent feature. The GAN inversion-based face swap methods [31, 53, 54] continue to demonstrate good performance. .0We also employ the style latent features of GAN inversion models to represent the temporal changes of facial attributes. Given that most of the recent GAN inversion models, including those based on the StyleGAN architecture [20], utilize style features, a similar situation applies to the construction of deepfake generation models. Hence, we propose the use of StyleGRU, which is well suited for countering attacks on state-of-the-art deepfake detection models.

3. Methodology

In this section, we propose the video deepfake detection framework that exploits the style latent vectors and captures their characteristics in the generated videos. Especially, we proposed four distinct modules to achieve the goal: (1) StyleGRU module that extracts the pSp features for each individual frame and encode the temporal variations of the style latent vectors using the GRU into the style-based temporal feature E_{style}, (2) 3D ResNet-50 module that encodes the video clip into the content feature C_{content}, (3) Style Attention Module (SAM) that combines the style-based temporal feature E_{style} with the content feature C_{content} via the attention mechanism, (4) Temporal Transformer Encoder (TTE) module that effectively maps the combined features into the binary classes (i.e. real and fake). Fig. 2 illustrates the overall flow of our framework.

3.1. Overall Architecture

3.1.1 Input configuration

We obtain the video clip I that contains a sequence of l consecutive frames extracted from a single video. Then, we build a clip I_p composed of all cropped faces within the input clip I that have been preprocessed with FFHQ [20] conditions. The preprocessed video clip I_p is used as an input to StyleGRU after resizing it to be $\mathcal{R}^{256\times256\times l}$, while the raw clip I is an input of 3D ResNet-50 module with the resized shape of $\mathcal{R}^{224\times224\times l}$.

3.1.2 StyleGRU module

The preprocessed video clip I_p serves as the input to extract style latent vectors via the pSp encoder [38]. The pSp encoder (pSp) provides the capability to extract style latent vectors for inverting given images within a well-trained StyleGAN [21] latent space. We extract the style latent vectors $\mathbf{S}$ as:

$$\mathbf{S} = pSp(I_p) \equiv \{s_1, s_2, ..., s_l\}, \tag{1}$$

where $s_k \in \mathcal{R}^{\mathcal{D}}$ is the D-dimensional style vector of the k-th frame of the clip I_p, $k \in [1, l]$, and l is the number of frames in clip.

The style latent vector that we extract through the pSp encoder is a feature for the GAN inversion task, so it is insufficient for direct application to the deepfake detection task. To enable the extension of style from the image domain to the video, we employ the difference of subsequent

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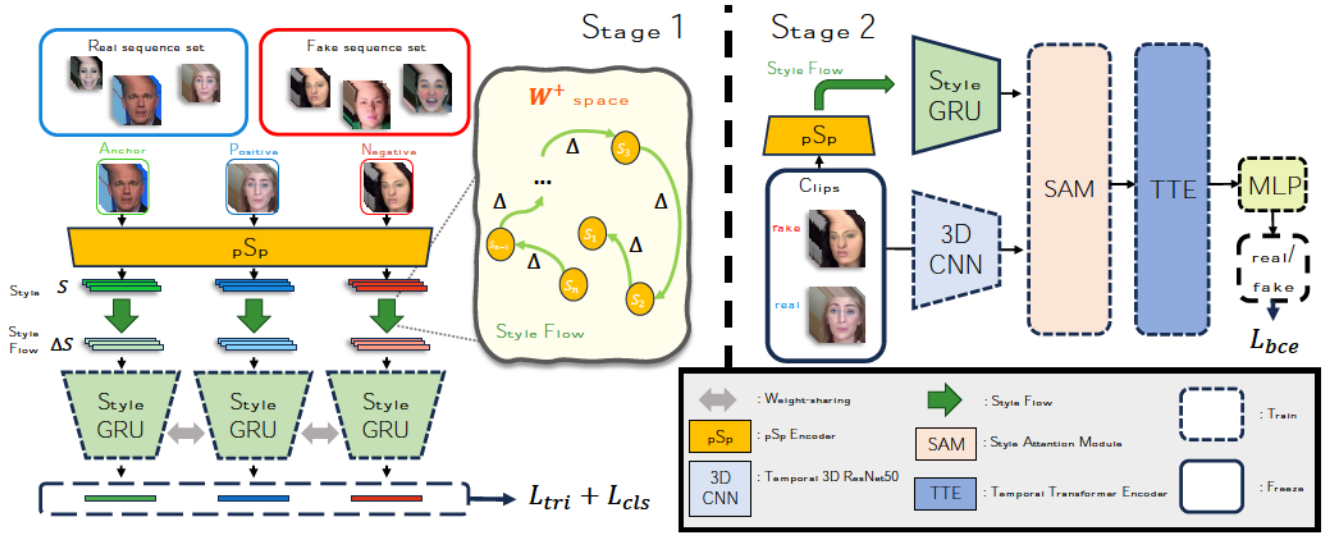


Figure 3. **Our training procedures.** In stage 1, we train the StyleGRU module using the supervised contrastive learning technique to effectively capture variations of the pSp feature and encode it into the robust style-based temporal feature. In stage 2, we train Style Attention Module (SAM) and Temporal Transformer Encoder (TTE) to integrate the style-based temporal feature and content feature and then map it towards the binary classes (ie. real and fake), respectively.

style latent vectors, extracted from frames within a clip. The style flow Δs, derived from difference of style latent vectors, is represented as $\Delta s_i = s_{i+1} - s_i$. Then, the input $\Delta \mathbf{S}$, which is called *style flow*, is constructed as follows:

$$\Delta \mathbf{S} = \{\Delta s_1, \Delta s_2, ..., \Delta s_{(l-1)}\}, \tag{2}$$

The style flow $\Delta \mathbf{S}$ is an input to the Gated Recurrent Unit (GRU) layers [7]. Then, we obtain a temporal embedded style flow $E_{\text{style}} \in \mathcal{R}^{C \times B \times L}$ through the hidden state of the final GRU layer. C represents the hidden unit size of GRU, B indicates the number of directions of GRU, and L denotes the number of GRU layers. We build two consecutive GRU layers of 4096 hidden units with bidirectional configuration, so $C = 4096$, $B = 2$, and $L = 2$. Thus, we can derive that:

$$E_{\text{style}} = \mathbf{GRU}(\Delta \mathbf{S}), \tag{3}$$

where $\mathbf{GRU}$ represents the operation of bidirectional two GRU layers embedded in StyleGRU module.

3.1.3 3D ResNet-50 module

The content feature C_{content} is derived from the feature extraction performed by the 3D CNN as:

$$C_{\text{content}} = \mathbf{3D\ CNN}(I), \tag{4}$$

where $\mathbf{3D\ CNN}$ is an operation function for extracting a feature from 3D CNN model. Thus, C_{content} contains information about visual and temporal inconsistency. We utilize the feature of $\mathcal{R}^{1024 \times 16}$ by applying a global average pooling to the last convolution layer of 3D-R50 architecture [3] pre-trained in FTCN [57].

3.1.4 Style attention module

We introduce a Style Attention Module (SAM) that leverages the style embedding vector obtained from the StyleGRU. To apply the attention mechanism, we designate C_{content} as the query, while E_{style} is allocated as the key and value. We employ the content feature as the query based on the intuition that these features hold the potential to identify anomalies in the style flow, driven by their encoded temporal inconsistency.

Thus, we design the query, key, and value of SAM as:

$$Q_C = \phi_q(C_{\text{content}}),\ K_E = \phi_k(E_{\text{style}}),\ V_E = \phi_v(E_{\text{style}}), \tag{5}$$

where ϕ_q, ϕ_k, and ϕ_v are linear projection function using fully connected layer to estimate the query, key, and value, respectively. Then, we calculate a Style Attention (SA) using the query and key as:

$$\mathbf{SA}(Q_C, K_E) = s\left(\frac{Q_C K_E}{\sqrt{d_{K_E}}}\right), \tag{6}$$

where $s(\bullet)$ represents a softmax function.

Ultimately, we compute the result of SAM by taking the dot-product of the SA with the value component. Thus,

$$\mathbf{SAM}(E, C) = \phi\big(\mathbf{SA}(Q_C, K_E) \cdot V_E\big), \tag{7}$$

where ϕ is the last linear projection layer of SAM.

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3.1.5 Temporal transformer encoder module.

The output from the SAM is linearly projected into 16 time steps (n=16) and then fed into the Temporal Transformer Encoder (TTE) module. The core of TTE is composed of standard transformer encoder blocks [49], with each block including a multi-head self-attention module and multi-layer perceptrons. The results of TTE are the input to the following classification head, yielding the final deepfake prediction through a sigmoid function σ. Thus,

$$\hat{y} = \sigma\big(f_{cls}(\mathbf{TTE}(C_{\text{content}} + \mathbf{SAM}(E_{\text{style}}, C_{\text{content}})))\big), \quad (8)$$

where f_{cls} is a function of fully-connected layer to represent the classification head. We refer to FTCN [57] for the architecture of TTE.

3.2. Training Procedure

The training procedure consists of two following stages. The first stage involves training a Gated Recurrent Unit (GRU) layers to enable encoding of the temporal style latent flow suitable for deepfake detection. In the following stage, we train the 3D CNN model for content embedding, the style attention module, and the subsequent modules for the binary classification. Fig. 3 provides a comprehensive illustration of our training procedures.

3.2.1 Stage 1: style representation learning

In stage 1, we train StyleGRU to learn the representation of style flow. We employ supervised contrastive learning [23] to produce an effective representation of style flow. For the supervised contrastive learning, we organized the style flows used as inputs to the GRU layers into three categories: anchor, positive, and negative style flows.

We utilize a different sampling strategy to build the clip for anchor, positive, and negative style flows. At first, the anchor style flow ($\Delta\mathbf{S}_a$) is extracted from the clip sampled from the video sequence by a sliding window sampling strategy. On the other hand, the positive style flow ($\Delta\mathbf{S}_p$) is obtained by using a random sampling strategy. Both the anchor and the positive style flows are extracted from the preprocessed clips with the same label. In contrast, the negative style flow ($\Delta\mathbf{S}_n$) is obtained from the clips with labels distinct from those of the anchor and positive clips. The negative style flow is generated by randomly employing one of the two sampling strategies.

As a batch, the anchor, positive, and negative style flows are fed into the GRU layers, and their representation is trained by using a triplet loss and a classification loss. When we define the GRU's final hidden states from $\Delta\mathbf{S}_a$, $\Delta\mathbf{S}_p$, and $\Delta\mathbf{S}_n$ by E_a, E_p, and E_n, respectively, the triplet loss for training the GRU layers is derived as follows:

$$L_{tri} = \mathbf{max}\big(||E_a - E_p||_2^2 - ||E_a - E_n||_2^2 + \alpha, 0\big), \quad (9)$$

where α is a margin parameter.

Additionally, we used classification loss (L_{cls}), which is designed by Binary Cross-Entropy (BCE) loss to learn appropriate features for classification. To obtain the predicted value $\hat{y}$ for the ground truth label y, the hidden states E of the GRU are input into the auxiliary classifier. The calculated BCE(y, $\hat{y}$) loss is utilized as L_{cls}. The auxiliary classifier consists of three fully connected layers, with a residual connection applied after the first two layers. The hidden size of each fully connected layer is kept identical at 4096.

Then, the entire loss function for training StyleGRU is as follows:

$$L = L_{tri} + \lambda L_{cls}, \quad (10)$$

where λ is a user-defined hyperparameter to determine the influence degree of L_{cls}. We minimize the loss by using Adam optimizer [25].

3.2.2 Stage 2: style attention-based deepfake detection.

In stage 2, to perform the real/fake binary classification task, we employ a conventional Binary Cross-Entropy (BCE) loss to train the 3D ResNet-50, SAM, and TTE modules, while fixing the StyleGRU module trained at stage 1. The BCE loss (L_{bce}) is applied to reduce the gap between the fake prediction $\hat{y}$ of Eq. 8 and its corresponding ground truth label, where 0 is real and 1 is fake. We used the momentum-driven SGD optimizer for the training process of stage 2.

4. Experiments

We conduct experiments in this section to demonstrate the superiority of our algorithm. Experiments under cross-dataset and cross-manipulation conditions serve as benchmark settings to assess the generalization performance of deepfake detection models.

Implementation Details. During the preprocessing phase, we utilized Dlib [24] to extract facial landmarks and perform alignment, as well as Retinaface [10] to carry out face cropping. Following this preprocessing, the real and fake clips fed into the model consist of a total of $l = 32$ frames each. In stage 1, a dropout layer with a dropout rate of $p = 0.2$ is included between the input style flow and the StyleGRU. Furthermore, in StyleGRU, an RNN dropout rate of $p = 0.1$ is applied. As training parameters, a total of 50,000 samples are organized with a batch size of 256 over 100 epochs for contrastive learning. For optimization, the ADAM optimizer is used with a learning rate of $5e^{-4}$.

In stage 2, we use the momentum SGD optimizer and set the weight decay to $1e^{-4}$. The training is conducted with a batch size of 16. For the first 10 epochs, the learning rate of SGD gradually warms up from 0.01 to 0.1. Subsequently, a cosine learning rate scheduler is used to decrease the learning rate throughout 100 epochs.

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Method	CDF	DFD	FSh	DFo	Avg
Xception [8]	73.7	-	72.0	84.5	-
CNN-aug [50]	75.6	-	65.7	74.7	-
PatchForensics [4]	69.6	-	57.8	81.8	-
Multi-task [35]	75.7	-	66.0	77.7	-
FWA [28]	69.5	-	65.5	50.2	-
Two-branch [34]	76.7	-	-	-	-
DCL [44]	82.3	91.6	-	-	-
Face X-ray [27]	79.5	95.4	92.8	86.8	88.6
NoiseDF [51]	75.9	-	-	70.9	-
SLADD [5]	79.7	-	-	-	-
SBI-R50 [43]	85.7	94.0	78.2	91.4	87.3
CNN-GRU [41]	69.8	-	80.8	74.1	-
STIL [15]	75.6	-	-	-	-
LipForensics [16]	82.4	-	97.1	97.6	-
RealForensics* [17]	85.6	82.2	**99.6**	**99.8**	91.8
FTCN* [57]	86.9	94.4	98.8	98.8	94.7
AltFreeze* [52]	**89.0**	93.7	99.2	99.0	95.2
StyleGRU	62.5	72.2	90.7	77.7	79.8
Ours	**89.0**	**96.1**	99.0	99.0	**95.8**

Table 1. **Generalization to cross-dataset.** We present the AUC scores (%) on the CDF, DFD, FSh, and DFo datasets to assess the cross-dataset performance of our framework. The highest-performing score is marked in **bold**. The underlined values correspond to the Top-2 best methods. The asterisk (*) denotes that we have reproduced the results using officially provided weights.The results of other methods were obtained from AltFreeze [52].

Datasets and Measurement. To ensure robustness under various conditions, we used the following deepfake detection datasets: (1) **FaceForensics++** (FF++) [40] comprises 1,000 original videos and 4,000 fake videos produced in four manipulation methods, Deepfake (DF) [9], Face2Face (F2F) [46], FaceSwap (FS) [32], NeuralTexture (NT) [47], and we utilize the c23 version. (2) **FaceShfiter** (FSh) [26] and (3) **DeeperForensics** [19] result from the employment of optimized manipulation techniques to the original videos of FF++. We executed our experiments using test videos adhering to the same split as in FF++. (4) **CelebDF-v2** (CDF) [29] is composed of 590 real videos and 5,639 fake videos. For our study, we engaged a subset of 518 test videos. (5) **DeepfakeDetection** (DFD) [13] consists of 363 real videos and 3071 synthetic videos.

To train our model, we used the training videos from the FF++ [40] dataset with light compression, specifically identified as c23. The evaluation metric is the Area Under the receiver operating characteristic Curve (AUC). Most deepfake detection algorithms have utilized AUC as their evaluation metric, and Lipforensics [16], which is a fundamental benchmark for video-based deepfake detection, also employed AUC.

Comparisons. We have compared both image-based and video-based deepfake detection models. Xception [40] is a method of training the Xception model [8], which is a widely used CNN architecture. CNN-aug [50] discovered that CNN-generated images can be readily identified by a CNN model. PatchForensics [4] indicated that the use of patch-based classifiers can contribute to robust deepfake detection. Multi-task [35] approach employed an architecture similar to that of an auto-encoder for deepfake detection. FWA [28] introduced a training data generation technique that relies on deteriorating the quality of the GAN-generated source images. Two-branch [34] method utilized multi-task learning within the context of a deepfake detection dataset. SLADD [5] introduced an algorithm that automatically learns effective augmentation techniques for deepfake detection. SBI [43] proposed a novel approach to blending artifacts using self-images. DCL [44] applied contrastive learning to the task of deepfake detection in the image domain. NoiseDF [51] compared the features of the cropped face and background squares using noise to detect deepfakes. Face X-ray [27] identified face forgery by segmenting and delineating the blending border between source and target images. CNN-GRU [41] highlighted the potential of utilizing CNN and GRU to investigate temporal features. STIL [15] specifically focused on learning spatio-temporal inconsistency. LipForensics [16] exhibited robust generalization performance through the use of a trained model to capture the representation of natural mouth movement. RealForensics [17] improved the ability to generalize using multimodal architecture. FTCN [57] argued that leveraging temporal inconsistency is beneficial for enhancing the generalizability of deepfake detection algorithms. AltFreeze [52] is an architecture that demonstrates good generalization performance by separately training spatial information and temporal information.

4.1. Generalization to cross-datasets

In the context of our framework, we have set the cross-dataset performance as a proxy for the generalizability in real-world situations. We train the proposed framework on the FF++ and then evaluate its performance on the CDF, DFD, FSh, and DFo datasets. Table 1 shows that our framework exhibits the best detection performance in the CDF, DFD, and DFo datasets, which include high-quality deepfake videos. The performance on the FSh dataset also reaches a top-2 level, with the highest average score, indicating that our model is a deepfake detection algorithm with generalization capability. 'StyleGRU' represents the cross-dataset score using the classification layer directly connected after the StyleGRU for the Stage 1 training. The score of StyleGRU is superior to that of CNN-GRU [41], which is a similar structure that utilizes image features, on the FSh and DFo datasets. This suggests the potential for a better impact on generalization performance by utilizing the flow of style latent vectors rather than simply relying on the image-based feature.

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Method	Train on remaining three				
	DF	FS	F2F	NT	Avg
Xception [8]	93.9	51.2	86.8	79.7	77.9
CNN-aug [50]	87.5	56.3	80.1	67.8	72.9
PatchForensics [4]	94.0	60.5	87.3	84.8	81.6
Face X-ray [27]	99.5	93.2	94.5	92.5	94.9
CNN-GRU [41]	97.6	47.6	85.8	86.6	79.4
LipForensics [16]	99.7	90.1	98.8	98.3	97.1
FTCN* [57]	99.8	99.3	95.9	95.3	97.5
AltFreeze [52]	99.8	99.7	98.6	96.2	98.6
StyleGRU	94.0	68.5	88.8	80.6	83.0
Ours	99.7	98.8	98.6	96.4	98.4

Table 2. **Generalization to cross-manipulation.** In order to quantitatively assess the performance of our method in cross-manipulation scenarios, we utilized video-level AUC. The asterisk (*) denotes results that we have reproduced. The underlined values correspond to the Top-3 best methods. The results of other methods were obtained from AltFreeze [52].

Method				Dataset		
StyleGRU	SAM	TTE	Δ	CDF	FSh	Avg
✓	-	-	✓	62.5	90.7	76.6
-	✓	✓	✓	85.7	98.4	92.1
✓	-	✓	✓	87.8	98.3	93.1
✓	✓	✓	-	88.4	98.4	93.4
✓	✓	✓	✓	89.0	99.0	94.0

Table 3. **Framework ablation.** We evaluate our framework on the CDF and FSh datasets using video-level AUC scores and conduct ablation studies for each stage of the proposed methodology. All models utilized the FF++ dataset for training.

Loss		Dataset		
L_{cls}	L_{tri}	FF++	CDF	FSh
✓	-	99.2	88.6	99.0
-	✓	98.8	88.4	98.8
✓	✓	99.1	89.0	99.0

Table 4. **Stage 1 loss ablation.** We conduct an ablation study for the losses employed in stage 1. Each experiment uses the different combinations of training losses for StyleGRU, while preserving the training process of stage 2. The model is trained by the FF++ dataset, and evaluated by video-level AUC.

4.2. Generalization to cross-manipulations

For deepfake detection tasks, the evaluation of cross-manipulation performance targets fake videos generated from the same original video but using a different scheme. A common approach is to select three out of the four manipulations provided by FF++, and then the model trained by the three chosen manipulation sets is evaluated by one remaining manipulation set. DF [9] and FS [32] represent identity swap manipulation methods, while F2F [46] and NT [47] denote expression swap manipulation methods. AltFreeze [52] does not officially provide weights for this experiment setting. Therefore, it is not possible for us to reproduce. Table 2 presents a comparison of the performance of the cross-manipulation between our model and various models. For expression swap manipulation methods like F2F and NT, they exhibit relatively lower performance compared to Lipforensics [16]. Expression swap inherently lead to inconsistencies in the mouth region, which is why LipForensics appears to perform well. However, our framework displays outstanding performance across all manipulations.

4.3. Ablation Study

In this section, we conduct ablation studies to verify the approach used in our framework. we validate the necessity of the proposed StyleGRU and SAM and conduct an ablation study on the losses used in Stage 1.

4.3.1 Component-wise Ablation Study

As shown in Table 3, we conducted experiments using only the StyleGRU by feeding the hidden states of StyleGRU into an MLP classification head to detect the real/fake of input video. Even though only the classification through StyleGRU was conducted, ours still demonstrates high scores on the Fsh dataset. In addition, considering the significant performance drop when StyleGRU is not used, it can be observed that our proposed StyleGRU contributes to improving cross-dataset performance. The influence of the proposed SAM in Stage 2 on the outcome is also evaluated. For experiments without StyleGRU, we processed style latent vectors extracted from clips using an MLP to match the dimension of the 3D CNN features. In the case of experiments without the SAM, we simply added the StyleGRU feature to the 3D CNN feature. The absence of the SAM during the blending of content and style features leads to a substantial decline in performance. This quantitatively demonstrates the efficacy of the SAM trained during stage 2 in achieving the adapted feature for robust deepfake detection. Finally, the influence of differentiating (Δ) the style latent vectors is investigated by replacing the input of the style flow with the original style latent vectors. During the overall stage 1 training process, it was observed that the lack of differencing in style latent vectors as a pre-processing to be fed into the StyleGRU leads to a slower decrease in the loss. Training without differentiation was confirmed to not only lead to slower progression in stage 1 learning but also adversely affect cross-dataset performance.

4.3.2 Ablation Study for Losses

Ablation studies are conducted on various options of training losses that can be employed at stage 1 training for StyleGRU representation learning. As shown in Table 4, when

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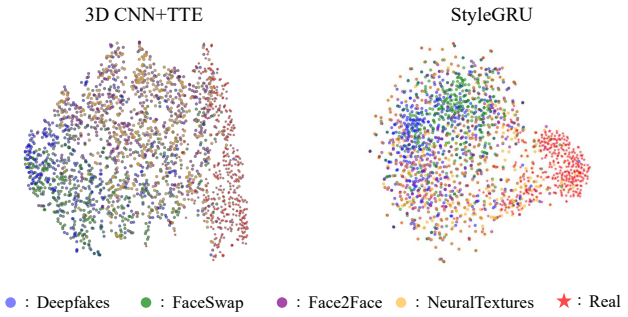


Figure 4. **t-SNE visualization.** The t-SNE visualization was conducted using the final layer features of a model trained on the FF++ test set. For hyperparameters, we utilized 1000 iterations with a perplexity of 40 and PCA to 30.

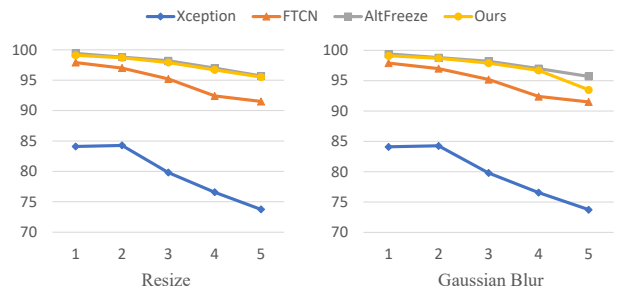


Figure 5. **Robustness to Unseen Perturbation.** The performance changes based on the video-level AUC metric when applying two perturbations at five different degradation levels. The perturbation follows the approach provided by DeeperForensics [19].

only classification loss (L_{cls}) is used, our model exhibits best performance on the training dataset, while suffering from the reduction of domain generality. Using only triplet loss (L_{tri}) results in relatively reduced performance due to the lack of distinctiveness between the real and fake videos for classification. Although employing both L_{cls} and L_{tri} results in a slight performance decrease on the seen dataset, we can see that their concurrent employment proves to be beneficial on the unseen datasets.

4.4. Analytic Experiments

In this section, we perform visualizations for further analysis of the proposed style latent flow in our study. We also analyze the perturbation robustness of our algorithm, which is important to consider in real-world scenarios.

4.4.1 t-SNE for Domain Generalization

To evaluate the learning performance of StyleGRU, we conduct t-SNE visualizations on the FF++ test set using a model trained on the FF++ dataset. As illustrated in Fig. 4, despite StyleGRU utilizing significantly fewer parameters, our model manages to extract more distinct features in the feature space. This characteristic represents our intuition that the distinctiveness of style latent vectors aligns well with the generality of deepfake video detection tasks across various generation algorithms.

4.4.2 Robustness to Unseen Perturbation

To evaluate the robustness of perturbation, experiments under the conditions proposed by DFo [19] are conducted to observe performance changes. In Fig. 5, performance under specific perturbation conditions exhibits a slight deficiency, which is attributed to the fact that the pSp encoder [38] used for extracting style latent vectors was created without considering noise. However, our model demonstrates a high level of robustness against most perturbations.

5. Conclusion

In this paper, we first demonstrate that the style latent flow extracted from consecutive frames of a video, serves as a effective cue for deepfake detection. Additionally, we confirm that leveraging an attention mechanism based on style flow facilitates model generalization. The style flow feature is trained via a supervised-contrastive representation learning process, and we develop a classification model using the Style Attention Module. The superior performance of the proposed algorithm validates the importance of temporal changes in style latent vectors to generalize the deepfake video detection task. While this paper focused on the facial attributes extracted by StyleGAN, our future plans is to broaden the scope of attributes for deepfake video detection to encompass various objects. We anticipate that utilizing StyleGAN pre-trained on other subjects, such as images of dogs and scenery, could broaden the applicability of our method to a wider range of deepfake subjects.

The model we propose includes an additional step of extracting style latent vectors through the pSp encoder. As a result, the data preprocessing takes a relatively long time. However, we confirms the positive impact of using style flow on the generalization performance of deepfake video detection models and provides a new direction for addressing synthesized videos created through GAN inversion.

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References

[1] Yuval Alaluf, Or Patashnik, and Daniel Cohen-Or. Restyle: A residual-based stylegan encoder via iterative refinement. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 6711–6720, 2021. 2

[2] Weiming Bai, Yufan Liu, Zhipeng Zhang, Bing Li, and Weiming Hu. Aunet: Learning relations between action units for face forgery detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 24709–24719, 2023. 2

[3] Joao Carreira and Andrew Zisserman. Quo vadis, action recognition? a new model and the kinetics dataset. In *proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 6299–6308, 2017. 4

[4] Lucy Chai, David Bau, Ser-Nam Lim, and Phillip Isola. What makes fake images detectable? understanding properties that generalize. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXVI 16*, pages 103–120. Springer, 2020. 6, 7

[5] Liang Chen, Yong Zhang, Yibing Song, Lingqiao Liu, and Jue Wang. Self-supervised learning of adversarial example: Towards good generalizations for deepfake detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 18710–18719, 2022. 2, 6

[6] Shen Chen, Taiping Yao, Yang Chen, Shouhong Ding, Jilin Li, and Rongrong Ji. Local relation learning for face forgery detection. In *Proceedings of the AAAI conference on artificial intelligence*, pages 1081–1088, 2021. 2

[7] Kyunghyun Cho, Bart Van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using rnn encoder-decoder for statistical machine translation. *arXiv preprint arXiv:1406.1078*, 2014. 4

[8] François Chollet. Xception: Deep learning with depthwise separable convolutions. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1251–1258, 2017. 6, 7

[9] Deepfakes. faceswap. https://github.com/deepfakes/faceswap, 2019. [Accessed 27-07-2023]. 6, 7

[10] Jiankang Deng, Jia Guo, Evangelos Ververas, Irene Kotsia, and Stefanos Zafeiriou. Retinaface: Single-shot multi-level face localisation in the wild. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 5203–5212, 2020. 5

[11] Prafulla Dhariwal and Alexander Nichol. Diffusion models beat gans on image synthesis. *Advances in neural information processing systems*, 34:8780–8794, 2021. 1

[12] Shichao Dong, Jin Wang, Renhe Ji, Jiajun Liang, Haoqiang Fan, and Zheng Ge. Implicit identity leakage: The stumbling block to improving deepfake detection generalization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3994–4004, 2023. 2

[13] Nick Dufour and Andrew Gully. Contributing Data to Deepfake Detection Research — ai.googleblog.com. https://ai.googleblog.com/2019/09/contributing-data-to-deepfake-detection.html, 2019. [Accessed 30-07-2023]. 6

[14] Qiqi Gu, Shen Chen, Taiping Yao, Yang Chen, Shouhong Ding, and Ran Yi. Exploiting fine-grained face forgery clues via progressive enhancement learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 735–743, 2022. 1, 2

[15] Zhihao Gu, Yang Chen, Taiping Yao, Shouhong Ding, Jilin Li, Feiyue Huang, and Lizhuang Ma. Spatiotemporal inconsistency learning for deepfake video detection. In *Proceedings of the 29th ACM international conference on multimedia*, pages 3473–3481, 2021. 1, 6

[16] Alexandros Haliassos, Konstantinos Vougioukas, Stavros Petridis, and Maja Pantic. Lips don't lie: A generalisable and robust approach to face forgery detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5039–5049, 2021. 1, 2, 6, 7

[17] Alexandros Haliassos, Rodrigo Mira, Stavros Petridis, and Maja Pantic. Leveraging real talking faces via self-supervision for robust forgery detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14950–14962, 2022. 6

[18] Yonghyun Jeong, Doyeon Kim, Youngmin Ro, and Jongwon Choi. Frepgan: robust deepfake detection using frequency-level perturbations. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 1060–1068, 2022. 2

[19] Liming Jiang, Ren Li, Wayne Wu, Chen Qian, and Chen Change Loy. Deeperforensics-1.0: A large-scale dataset for real-world face forgery detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 2889–2898, 2020. 1, 6, 8

[20] Tero Karras, Samuli Laine, and Timo Aila. A style-based generator architecture for generative adversarial networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4401–4410, 2019. 2, 3

[21] Tero Karras, Samuli Laine, Miika Aittala, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. Analyzing and improving the image quality of stylegan. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 8110–8119, 2020. 2, 3

[22] Tero Karras, Miika Aittala, Samuli Laine, Erik Härkönen, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. Alias-free generative adversarial networks. *Advances in Neural Information Processing Systems*, 34:852–863, 2021. 2

[23] Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. Supervised contrastive learning. *Advances in neural information processing systems*, 33:18661–18673, 2020. 2, 5

[24] Davis E King. Dlib-ml: A machine learning toolkit. *The Journal of Machine Learning Research*, 10:1755–1758, 2009. 5

[25] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014. 5

[26] Lingzhi Li, Jianmin Bao, Hao Yang, Dong Chen, and Fang Wen. Faceshifter: Towards high fidelity and occlusion aware face swapping. *arXiv preprint arXiv:1912.13457*, 2019. 6

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[27] Lingzhi Li, Jianmin Bao, Ting Zhang, Hao Yang, Dong Chen, Fang Wen, and Baining Guo. Face x-ray for more general face forgery detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 5001–5010, 2020. 2, 6, 7

[28] Yuezun Li and Siwei Lyu. Exposing deepfake videos by detecting face warping artifacts. arxiv 2018. *arXiv preprint arXiv:1811.00656*, 2018. 1, 6

[29] Yuezun Li, Xin Yang, Pu Sun, Honggang Qi, and Siwei Lyu. Celeb-df: A large-scale challenging dataset for deepfake forensics. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3207–3216, 2020. 1, 6

[30] Zhengzhe Liu, Xiaojuan Qi, and Philip H.S. Torr. Global texture enhancement for fake face detection in the wild. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020. 1

[31] Zhian Liu, Maomao Li, Yong Zhang, Cairong Wang, Qi Zhang, Jue Wang, and Yongwei Nie. Fine-grained face swapping via regional gan inversion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8578–8587, 2023. 3

[32] MarekKowalski. Faceswap. https://github.com/MarekKowalski/FaceSwap, 2019. [Accessed 30-07-2023]. 6, 7

[33] Francesco Marra, Diego Gragnaniello, Luisa Verdoliva, and Giovanni Poggi. Do gans leave artificial fingerprints? In *2019 IEEE conference on multimedia information processing and retrieval (MIPR)*, pages 506–511. IEEE, 2019. 1

[34] Iacopo Masi, Aditya Killekar, Royston Marian Mascarenhas, Shenoy Pratik Gurudatt, and Wael AbdAlmageed. Two-branch recurrent network for isolating deepfakes in videos. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part VII 16*, pages 667–684. Springer, 2020. 6

[35] Huy H Nguyen, Fuming Fang, Junichi Yamagishi, and Isao Echizen. Multi-task learning for detecting and segmenting manipulated facial images and videos. In *2019 IEEE 10th international conference on biometrics theory, applications and systems (BTAS)*, pages 1–8. IEEE, 2019. 6

[36] Utkarsh Ojha, Yuheng Li, and Yong Jae Lee. Towards universal fake image detectors that generalize across generative models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 24480–24489, 2023. 1

[37] Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever. Zero-shot text-to-image generation. In *International Conference on Machine Learning*, pages 8821–8831. PMLR, 2021. 1

[38] Elad Richardson, Yuval Alaluf, Or Patashnik, Yotam Nitzan, Yaniv Azar, Stav Shapiro, and Daniel Cohen-Or. Encoding in style: a stylegan encoder for image-to-image translation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 2287–2296, 2021. 2, 3, 8

[39] Jonas Ricker, Simon Damm, Thorsten Holz, and Asja Fischer. Towards the detection of diffusion model deepfakes. *arXiv preprint arXiv:2210.14571*, 2022. 1

[40] Andreas Rossler, Davide Cozzolino, Luisa Verdoliva, Christian Riess, Justus Thies, and Matthias Nießner. Faceforensics++: Learning to detect manipulated facial images. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 1–11, 2019. 1, 2, 6

[41] Ekraam Sabir, Jiaxin Cheng, Ayush Jaiswal, Wael AbdAlmageed, Iacopo Masi, and Prem Natarajan. Recurrent convolutional strategies for face manipulation detection in videos. *Interfaces (GUI)*, 3(1):80–87, 2019. 1, 2, 6, 7

[42] Ekraam Sabir, Jiaxin Cheng, Ayush Jaiswal, Wael AbdAlmageed, Iacopo Masi, and Prem Natarajan. Recurrent convolutional strategies for face manipulation detection in videos. *Interfaces (GUI)*, 3(1):80–87, 2019. 2

[43] Kaede Shiohara and Toshihiko Yamasaki. Detecting deepfakes with self-blended images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 18720–18729, 2022. 2, 6

[44] Ke Sun, Taiping Yao, Shen Chen, Shouhong Ding, Jilin Li, and Rongrong Ji. Dual contrastive learning for general face forgery detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 2316–2324, 2022. 6

[45] Ayush Tewari, Mohamed Elgharib, Gaurav Bharaj, Florian Bernard, Hans-Peter Seidel, Patrick Pérez, Michael Zollhofer, and Christian Theobalt. Stylerig: Rigging stylegan for 3d control over portrait images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6142–6151, 2020. 3

[46] Justus Thies, Michael Zollhofer, Marc Stamminger, Christian Theobalt, and Matthias Nießner. Face2face: Real-time face capture and reenactment of rgb videos. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2387–2395, 2016. 6, 7

[47] Justus Thies, Michael Zollhöfer, and Matthias Nießner. Deferred neural rendering: Image synthesis using neural textures. *Acm Transactions on Graphics (TOG)*, 38(4):1–12, 2019. 6, 7

[48] Omer Tov, Yuval Alaluf, Yotam Nitzan, Or Patashnik, and Daniel Cohen-Or. Designing an encoder for stylegan image manipulation. *ACM Transactions on Graphics (TOG)*, 40(4): 1–14, 2021. 2

[49] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017. 5

[50] Sheng-Yu Wang, Oliver Wang, Richard Zhang, Andrew Owens, and Alexei A Efros. Cnn-generated images are surprisingly easy to spot... for now. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 8695–8704, 2020. 6, 7

[51] Tianyi Wang and Kam Pui Chow. Noise based deepfake detection via multi-head relative-interaction. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 14548–14556, 2023. 1, 6

[52] Zhendong Wang, Jianmin Bao, Wengang Zhou, Weilun Wang, and Houqiang Li. Altfreezing for more general video

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face forgery detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4129–4138, 2023. 2, 6, 7

[53] Chao Xu, Jiangning Zhang, Miao Hua, Qian He, Zili Yi, and Yong Liu. Region-aware face swapping. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7632–7641, 2022. 3

[54] Yangyang Xu, Bailin Deng, Junle Wang, Yanqing Jing, Jia Pan, and Shengfeng He. High-resolution face swapping via latent semantics disentanglement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7642–7651, 2022. 3

[55] Zhiyuan Yan, Yong Zhang, Yanbo Fan, and Baoyuan Wu. UCF: uncovering common features for generalizable deepfake detection. In *IEEE/CVF International Conference on Computer Vision, ICCV 2023, Paris, France, October 1-6, 2023*, pages 22355–22366. IEEE, 2023. 2

[56] Hanqing Zhao, Tianyi Wei, Wenbo Zhou, Weiming Zhang, Dongdong Chen, and Nenghai Yu. Multi-attentional deepfake detection. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2185–2194, 2021. 2

[57] Yinglin Zheng, Jianmin Bao, Dong Chen, Ming Zeng, and Fang Wen. Exploring temporal coherence for more general video face forgery detection. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 15044–15054, 2021. 1, 2, 4, 5, 6, 7

[58] Jiapeng Zhu, Yujun Shen, Deli Zhao, and Bolei Zhou. In-domain gan inversion for real image editing. In *European conference on computer vision*, pages 592–608. Springer, 2020. 2

[59] Jun-Yan Zhu, Philipp Krähenbühl, Eli Shechtman, and Alexei A Efros. Generative visual manipulation on the natural image manifold. In *Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11-14, 2016, Proceedings, Part V 14*, pages 597–613. Springer, 2016. 2

[60] Yuhao Zhu, Qi Li, Jian Wang, Cheng-Zhong Xu, and Zhenan Sun. One shot face swapping on megapixels. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4834–4844, 2021. 3

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Forgery-Aware Adaptive Learning With Vision Transformer for Generalized Face Forgery Detection

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***Abstract*— With the rapid progress of generative models, the current challenge in face forgery detection is how to effectively detect realistic manipulated faces from different unseen domains. Though previous studies show that pre-trained Vision Transformer (ViT) based models can achieve some promising results after fully fine-tuning on the Deepfake dataset, their generalization performances are still unsatisfactory. To this end, we present a Forgery-aware Adaptive Vision Transformer (FA-ViT) under the adaptive learning paradigm for generalized face forgery detection, where the parameters in the pre-trained ViT are kept fixed while the designed adaptive modules are optimized to capture forgery features. Specifically, a global adaptive module is designed to model long-range interactions among input tokens, which takes advantage of self-attention mechanism to mine global forgery clues. To further explore essential local forgery clues, a local adaptive module is proposed to expose local inconsistencies by enhancing the local contextual association. In addition, we introduce a fine-grained adaptive learning module that emphasizes the common compact representation of genuine faces through relationship learning in fine-grained pairs, driving these proposed adaptive modules to be aware of fine-grained forgery-aware information. Extensive experiments demonstrate that our FA-ViT achieves state-of-the-arts results in the cross-dataset evaluation, and enhances the robustness against unseen perturbations. Particularly, FA-ViT achieves 93.83% and 78.32% AUC scores on Celeb-DF and DFDC datasets in the cross-dataset evaluation. The code and trained model have been released at: https://github.com/LoveSiameseCat/FAViT.**

***Index Terms*— Face forgery detection, vision transformer, adaptive learning, generalization.**

I. Introduction

With the rapid development of deep learning technology, Artificial Intelligence-Generated Content (AIGC) technology has made significant progress in a wide variety of multimedia tasks. However, this advancement poses a grand challenge to the human eye in discriminating these digital contents. In particular, attackers can easily generate falsified facial content (a.k.a. *Deepfakes*) for various malicious purposes, posing a pressing threat to society over financial fraud, political conflict, and impersonation. To protect the authentication integrity, it is of paramount importance to develop effective methods for face forgery detection.

Most previous works build their detectors using Convolutional Neural Network (CNN), of which Xception [1] or EfficientNet [2] is widely employed as the basic backbone due to their outstanding Deepfake detection performance. To enhance the generalizability, some works explore the common forgery clues hidden in manipulated faces, such as noise information [3], [4], [5], blending artifacts [6], [7], [8], and frequency features [9], [10], [11], etc. However, the limited receptive field in CNNs constrains their capacity to comprehensively learn more generalized features [12]. In response, some methods seek to use Vision Transformers (ViT) for face forgery detection [13], [14], [15], [16]. Due to the self-attention mechanism, these ViT-based methods can model the long-range relations among different input tokens. However, ViT struggles to capture the local feature details, which are particularly crucial in Deepfake detection [17]. To address this limitation, previous works incorporate CNN local priors into ViT architectures [18], [19].

Pre-trained models are known to exhibit better convergence and generalization for downstream tasks as compared to models trained from scratch [20], and the pre-trained ViT[1] has been proven effective in forensic tasks [21]. Therefore, in previous works [13], [17], [18], [19], it is common to use publicly accessible pre-trained weights to initialize the ViT-based detectors, and subsequently update these parameters on Deepfake

[1]Specifically refers to models trained on public large-scale vision datasets.

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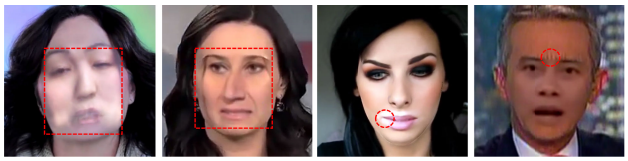


Fig. 1. Examples of forgery artifacts in manipulated faces that cause global inconsistency (marked with rectangular box) and local inconsistency (marked with circular box).

datasets. However, recent works [22], [23] point out that ViT-based models fully fine-tuned on specific down-stream tasks would disrupt pre-trained features and may overfit to data-specific patterns [21], potentially hindering their generalization capability in open-set environments. On the other hand, as shown in Fig. 1, forgery artifacts in manipulated faces can cause either global or local inconsistencies, indicating the need to model critical representations from multiple perspectives. Based on these observations, we propose the **F**orgery-aware **A**daptive **Vi**sion **T**ransformer (FA-ViT) for generalized face forgery detection, where the pre-trained weights are fixed, and only the designed adaptive modules are optimized to capture abundant forgery-aware information from both global and local perspectives.

The overview of FA-ViT is illustrated in Fig. 2. It is primarily composed of two modules: the Global Adaptive Module (GAM) and the Local Adaptive Module (LAM) to capture global and local forgery-aware information. Specifically, GAM is designed to receive ViT tokens and interact with the original query, key, and value outputs of the self-attention layer. Given that global artifacts span a large region of the manipulated face, GAM leverages the self-attention mechanism to enhance its ability to learn global forensics clues from these artifacts. On the other hand, LAM employs quadratic encoding [24] to enhance the local association between each ViT token and its neighboring spatial information, guiding the model in emphasizing the anomalies among local inconsistencies. As such, the global and local forgery-aware information jointly adapt to the fixed pre-trained features, forming the generalized forensic representation for detecting Deepfakes across various scenarios.

Furthermore, the commonly used cross-entropy loss emphasizes category-level differences but struggles to capture the fine-grained information that reveals subtle differences between manipulated and genuine faces [10], [25], [26]. To solve this issue, we design the Fine-grained Adaptive Learning (FAL). As shown in Fig. 2, the fine-grained pairs, which share similar visual semantic content but belong to different categories, are grouped together as input pairs. FAL utilizes the weight from the last Fully Connected (FC) layer as the proxy prototype of genuine faces, and regularizes the relationship between the prototype and each fine-grained pair through circle loss [27]. Under the guidance of FAL, the proposed adaptive modules are driven to mine more fine-grained forgery-aware information to compact the genuine faces in feature space, thus further improving model's generalizability.

Our main contributions are summarized as follows:

- We observe that ViT-based models struggle to generalize to unseen datasets when fully fine-tuning on the task of Deepfake detection. To address this issue, we propose the novel Forgery-Aware Adaptive Vision Transformer (FA-ViT) tailored for generalized face forgery detection under the adaptive learning paradigm.
- We propose the Global Adaptive Module (GAM) and the Local Adaptive Module (LAM), which effectively adapt global and local forgery-aware information to the pre-trained ViT features. In addition, we design a novel Fine-grained Adaptive Learning (FAL) to guide these adaptive models to capture fine-grained information in the adaptive learning process.
- We conduct extensive experiments on multiple datasets and senarios, and the results demonstrate our proposed FA-ViT outperforms state-of-the-art methods in a wide variety of evaluations.

II. Related Work

A. Face Forgery Detection

Early attempts in face forgery detection mainly relied on hand-crafted features, such as visual artifacts [28], face wrapping distortions [29], and abnormal blinking frequency [30]. However, these methods suffer from limited detection accuracy. Recent studies design neural networks for face forgery detection, like MesoNet [31], CapsuleNet [32], Xception [1], and EfficientNet [2]. However, these CNN-based methods are prone to overfitting to the training data, making them difficult to generalize well to unseen datasets. To improve the generalization performance, follow-up works [33], [34], [35], [36] delve into domain expert knowledge to guide the model in learning generalized representations. Face X-ray [6], SBI [8], and LAA-Net [7] build well-annotated faces in an unsupervised manner, but the learned blending artifacts can be easily eliminated by common perturbations. SCL [10], RECCE [37], DCL [25], Lisiam [38], and CFM [39] explore auxiliary information to learn more critical forgery features, while F3Net [11] and SPSL [40] reveal manipulation traces beyond RGB space. LSDA [41] diversifies forgery types by augmenting forgery features in the latent space. DSRL [42] develops a spatial-frequency reconstruction framework to capture diverse forgery-related features and enhances the learned representation using a dynamic filtering module. Although these local features extracted from CNN are effective in face forgery detection, the limited receptive field constrains their capability to learn global representations [43], [44].

In contrast, ViT-like architectures model the global relations through self-attention mechanism and capture long-distance information in the final representation. For instance, AUNet [45] explores the relationship between different facial action units using a transformer and detects manipulated faces based on the correlation intensity among these action units. Zhang *et. al.* [46] introduce a vision and language transformer-based framework to enhance the vision representations with multi-modal information. Due to the lack of inductive bias, ViT often ignore the local feature details, which are rather critical in face forgery detection [17], [47], [48]. This issue can be mitigated by introducing local prior knowledge into ViT. For example, UIA [18] supervises the

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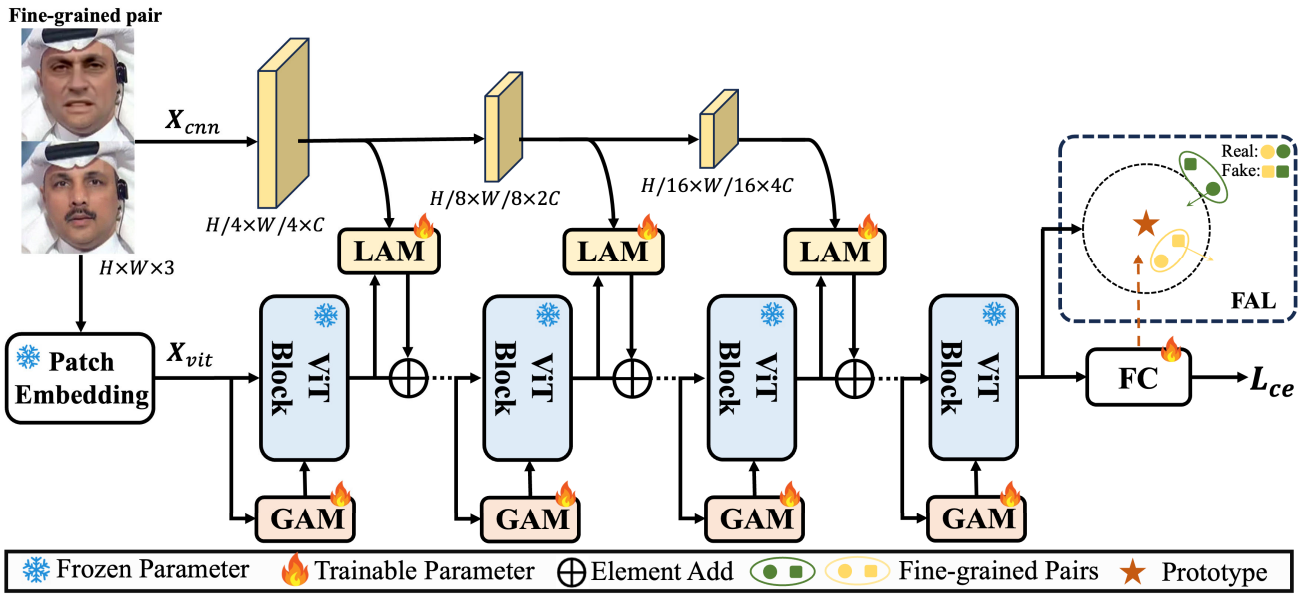


Fig. 2. Overview of the proposed **F**orgery-aware **A**daptive **Vi**sion **T**ransformer (FA-ViT). Global Adaptive Modules (GAM) and Local Adaptive Modules (LAM) are inserted into the fixed pre-trained ViT to learn different types of forgery-aware information. Fine-grained Adaptive Learning (FAL) further guides these modules to mine fine-grained clues from subtle discrepancies in fine-grained pairs. We omit CNN blocks for better visualization.

attention map generated from self-attention layers through local consistency learning. M2TR [14], HFI-Net [19], and F2trans [17] combine CNN and ViT features to learn complementary information in the final representation. However, these ViT-based methods are designed in the fully fine-tuning paradigm, which may lead to the loss of valuable pre-trained knowledge. To solve this, our proposed FA-ViT utilizes additional modules to adaptively learn forgery-aware information while keeping ViT parameters preserved, resulting in a better representation learning and enhanced detection performance.

B. Model Adaptation

In general, fine-tuning all model parameters is widely used for downstream tasks. However, recent works [22], [23] have demonstrated that the fully fine-tuning strategy may disrupt pre-trained features and does not generalize well to out-of-distribution data. As pre-trained models become larger and stronger, the efficiency and stability issues of the fully fine-tuning strategy become more severe. Ojha et al. [21] reveal that training only a learnable classifier for a fixed CLIP-ViT achieves better generalization performance than fully fine-tuning strategy in synthetic image detection. However, this is a sub-optimal solution since the fixed model lacks task-specific knowledge. To explore new model adaptation strategies, Lee et al. [49] propose an alternative strategy by selectively fine-tuning a subset of layers, which demonstrated a better performance for image corruption task. Nevertheless, manually selecting the optimal layers for fine-tuning is challenging and time-consuming. To address this issue, Side-Tuning [50], to the best of our knowledge, is the first work to explore model adaptation by using additive modules with fixed pre-trained models. It established a baseline and verified the effectiveness across different tasks. Subsequently, this adaptation strategy has been successfully applied to different tasks with specific expert design, including video action recognition [51], natural language understanding [52], semantic segmentation [53], and anti-spoofing [54]. Recently, HDDA [55] uses human perception salience to guide the model in learning essential features and introduces an adapter to reduce overfitting to less significant features. Shao *et. al.* [56] propose a dual-level adapter based detector to capture both global and local low-level manipulation traces for face forgery detection. Our experiments in Sec. IV-F.1 also demonstrated that fully fine-tuning the pre-trained ViT model leads to sub-optimal performance compared to other model adaptation strategies. Grounded in these observations, we customize a novel adaptive learning scheme on the pre-trained ViT backbone to achieve more generalized face forgery detection.

C. Fine-Grained Information Learning

As counterfeits become increasingly realistic, formulating face forgery detection as a fine-grained classification task [26], [47], [48] is preferable to expose local and subtle forgery traces. Zhao et al. [47] propose a multi-attentional learning framework that extracts local discriminative features from different attention maps. Lu et al. [26] further extend this attention mechanism into time domain to expose long distance inconsistencies. SFDG [57] explores content-adaptive features by fusing fine-grained and coarse-grained frequency information via dynamic graph learning. WATCHER [58] introduces a wavelet-guided texture-content hierarchical relation learning framework to explore the relation-aware information between fine-grained forgery clues and general visual representation. MTD-Net [59] and F2Trans [17] utilize the central difference operation [60] to reveal fine-grained forgery clues. The aforementioned works have demonstrated that fine-grained information is crucial for achieving generalized face forgery detection. In our proposed adaptive learning paradigm, FAL is proposed to drive these adaptive modules to focus on subtle discrepancies in each fine-grained pair, thereby facilitating the

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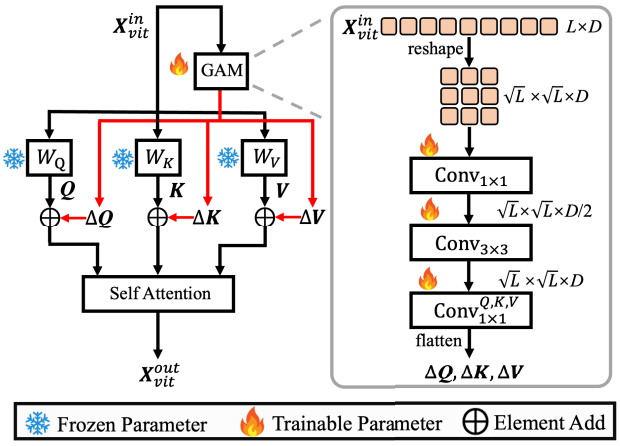


Fig. 3. The details of the proposed global adaptive module.

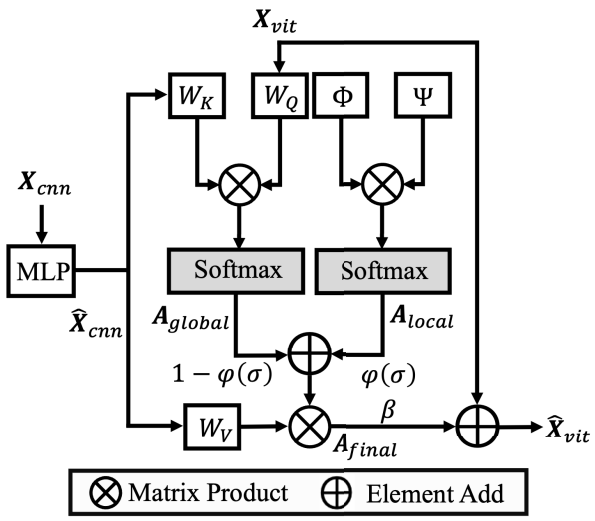


Fig. 4. The details of the proposed local adaptive module.

learning of fine-grained information from the perspective of the optimization objective.

III. Proposed Method

In this section, we first provide an overview of the proposed FA-ViT in Sec. III-A. Then we describe the Global Adaptive Module (GAM) and Local Adaptive Module (LAM) in Secs. III-B and III-C, respectively. Finally, the Fine-grained Adaptive Learning (FAL) is presented in Sec. III-D.

A. Overview

The framework of our proposed FA-ViT is illustrated in Fig. 2. It adopts the pre-trained ViT as the basic backbone, which comprises twelve ViT blocks. Each block consists of a Self-Attention (SA) layer and a Multi-Layer Perceptron (MLP) layer. During training, the parameters within these blocks remain frozen.

In FA-ViT, the input $\mathbf{X} \in \mathbb{R}^{H\times W\times 3}$ is divided into L image patches and further processed into D dimensional tokens, denoted as $\mathbf{X}_{vit} \in \mathbb{R}^{L\times D}$. In each ViT block, we insert a GAM into the SA layer. When $\mathbf{X}_{vit}$ passes through the SA layer, GAM captures the global forgery-aware information with the help of self-attention mechanism. On the other hand, we extract multi-scale spatial features from $\mathbf{X}$, where each scale feature $\mathbf{X}_{cnn}$ is obtained from a CNN block consisting of three convolutional layers. LAM aggregates spatial forgery-aware information from $\mathbf{X}_{cnn}$ for each ViT token, which benefits in capturing rich local details in an adaptive learning manner. In addition to the commonly used Cross Entropy (CE) loss, we introduce FAL to guide GAMs and LAMs to capture more fine-grained forgery-aware information.

B. Global Adaptive Module

The self-attention layer is a key component in ViT, which enables each input token to aggregate information from all other tokens. We first briefly introduce the calculation process in self-attention layer as follow.

Denote $\mathbf{X}_{vit}^{in} \in \mathbb{R}^{L\times D}$ is the input of the ViT block, it is first projected to query tokens $\mathbf{Q} \in \mathbb{R}^{L\times D}$, key tokens $\mathbf{K} \in \mathbb{R}^{L\times D}$ and value tokens $\mathbf{V} \in \mathbb{R}^{L\times D}$ through three learnable matrices $W_Q \in \mathbb{R}^{D\times D}$, $W_K \in \mathbb{R}^{D\times D}$ and $W_V \in \mathbb{R}^{D\times D}$:

$$\mathbf{Q} = \mathbf{X}_{vit}^{in} W_Q,\ \mathbf{K} = \mathbf{X}_{vit}^{in} W_K,\ \mathbf{V} = \mathbf{X}_{vit}^{in} W_V. \tag{1}$$

Then the calculation in self-attention layer is expressed as:

$$\mathbf{X}_{vit}^{out} = \text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}(\mathbf{Q}\mathbf{K}^{\top}/\sqrt{D})\mathbf{V}, \tag{2}$$

where $\mathbf{X}_{vit}^{out} \in \mathbb{R}^{L\times D}$ is the output.

The proposed GAM is built on the self-attention layer to mine global information with the self-attention mechanism. Previous works [18], [61], [62] have demonstrated that token embeddings (i.e., patch-wise representations) exhibit stronger correlations with their neighbors than with distant tokens. Therefore, convolutional layers with a bottleneck structure are adopted to model this local spatial relationship in GAM, introducing token-related local prior into the global adaptive learning process. The detail of our proposed GAM is shown in Fig. 3. Specifically, $\mathbf{X}_{vit}^{in}$ is first reshaped into the shape of $\sqrt{L} \times \sqrt{L} \times D$ according to its original spatial position. Then a 1×1 convolutional layer is applied to reduce the dimension, followed by a 3×3 convolutional layer to capture the token-level dependency. Finally, GAM generates global adaptive information for $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ through three different 1×1 convolutions. This process is formulated as follows:

$$\triangle\mathbf{Q}, \triangle\mathbf{K}, \triangle\mathbf{V} = \text{Conv}_{1\times 1}^{Q,K,V}(\text{Conv}_{3\times 3}(\text{Conv}_{1\times 1}(\mathbf{X}_{vit}^{in}))), \tag{3}$$

$$\mathbf{X}_{vit}^{out} = \text{Attention}(\mathbf{Q}+\triangle\mathbf{Q}, \mathbf{K}+\triangle\mathbf{K}, \mathbf{V}+\triangle\mathbf{V}), \tag{4}$$

where $\triangle\mathbf{Q}$, $\triangle\mathbf{K}$, and $\triangle\mathbf{V}$ are the adaptive information for original $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$. Since the original and the adaptive information are fused together and subsequently processed by the self-attention operation, this ensures that GAM interacts with all the tokens to capture forgery-aware features from a global perspective. It is worth noting that the parameters in $\text{Conv}_{1\times 1}^{Q,K,V}$ are initialized to zero, which contributes to stable learning of global forgery-aware knowledge.

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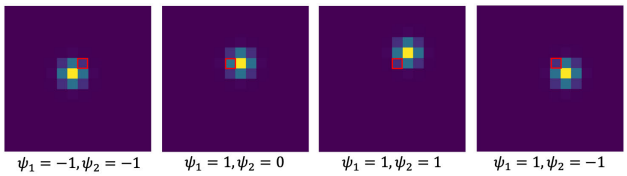


Fig. 5. The local attention map of the query token (marked by a red box) under different combination of ψ_1 and ψ_2.

C. Local Adaptive Module

The pure ViT processes input through stacked linear layers, which struggles to capture local details crucial for detecting manipulated faces. As shown in Fig. 1, artifacts in manipulated faces often introduce local inconsistencies, indicating that each query token must emphasize its relationships with surrounding positions to detect local forgery clues. Previous approaches use cross-attention modules [14], [15], [63] or addition operations [17], [64] to inject local spatial information into ViT-like architectures but often overlook the contextual importance of each query token. To alleviate this issue, our proposed LAM is designed to emphasize contextual information in the adaptive learning process, thus efficiently capturing critical local forgery clues from local inconsistencies.

Fig. 4 provides the details of our proposed LAM. The spatial feature $\mathbf{X}_{cnn}$ is first projected to $\hat{\mathbf{X}}_{cnn}$ through a MLP layer, where $\hat{\mathbf{X}}_{cnn}$ and $\mathbf{X}_{vit}$ share the same shape. For the ath token $\mathbf{X}^a_{token}$ in $\mathbf{X}_{vit}$, LAM calculates its attention score at different parts of $\hat{\mathbf{X}}_{cnn}$ by simultaneously considering its global attention $\mathbf{A}^a_{global}$ and local attention $\mathbf{A}^a_{local}$:

$$\mathbf{A}^a_{final} = (1 - \varphi(\sigma))\mathbf{A}^a_{global} + \varphi(\sigma)\mathbf{A}^a_{local}, \tag{5}$$

where $\mathbf{A}^a_{final}$ is the final attention map for $\mathbf{X}^a_{token}$. $\varphi(\cdot)$ represents sigmoid function, and σ is a learnable parameter with zero initialization. The calculation of $\mathbf{A}^a_{global}$ is similar to cross-attention, which is expressed as follows:

$$\mathbf{A}^a_{global} = softmax((\mathbf{X}^a_{token}W_Q)(\hat{\mathbf{X}}_{cnn}W_K)^T). \tag{6}$$

$\mathbf{A}^a_{local}$ is introduced to emphasize the locality of surrounding spatial information for $\mathbf{X}^a_{token}$ by using quadratic encoding [24]:

$$\mathbf{A}^a_{local} = softmax(\Phi\Psi^T), \tag{7}$$

where $\Phi \in \mathbb{R}^{\sqrt{L}\times\sqrt{L}\times 3}$ represents the locality strength, and $\Psi \in \mathbb{R}^{1\times 3}$ is the directional vector which determines the attention direction of $\mathbf{X}^a_{token}$. Note that we omit flatten operation before softmax operation for simplify. Supposing the spatial position of $\mathbf{X}^a_{token}$ is (i_a, j_a), and similarly, the spatial position of bth token in $\hat{\mathbf{X}}_{cnn}$ is (i_b, j_b). The relative locality strength $\phi_{a,b} \in \mathbb{R}^{1\times 3}$ in position (a, b) of Φ is expressed as:

$$\phi_{a,b} = \left(\|(i_b - i_a, j_b - j_a)\|_2, i_b - i_a, j_b - j_a\right). \tag{8}$$

On the other hand, the directional vector Ψ is expressed as:

$$\Psi = (-1, 2\psi_1, 2\psi_2), (\psi_1, \psi_2) \in \{-1, 0, 1\}. \tag{9}$$

In practice, we assign different values to ψ_1 and ψ_2 at different heads to explore local information in various directions, as illustrated in Fig. 5. We gather the final attention map for each query token to form $\mathbf{A}_{final}$ for $\mathbf{X}_{vit}$, and use it to inject local spatial information into ViT features. This process is formulated as follows:

$$\hat{\mathbf{X}}_{vit} = \mathbf{X}_{vit} + \beta\mathbf{A}_{final}\hat{\mathbf{X}}_{cnn}W_V, \tag{10}$$

where β is a learnable scaling factor with zero initialization, and $\hat{\mathbf{X}}_{vit}$ is passed to next ViT block. Empirically, the multi-scale spatial information are injected into the first, fourth and seventh ViT block, respectively.

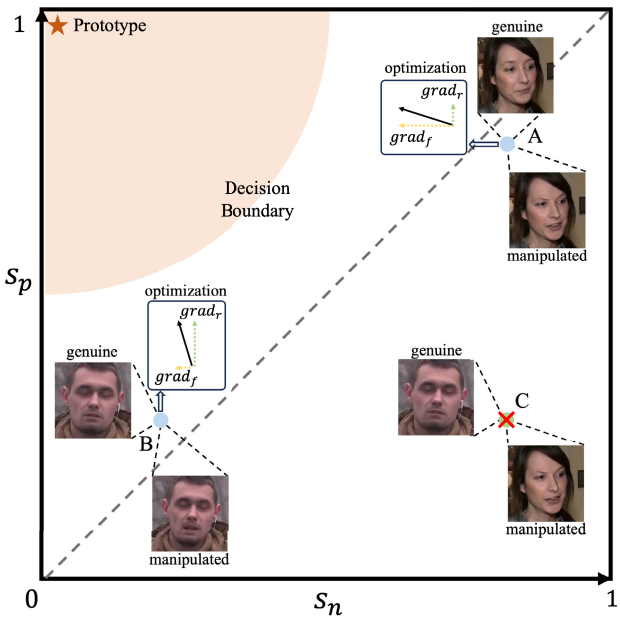


Fig. 6. Optimization in fine-grained adaptive learning. The weight vector corresponding to the genuine face in the FC layer is used as the prototype of genuine faces. $grad_r$ and $grad_f$ are the gradients from genuine (real) and manipulated (fake) faces, respectively. (Best viewed in color).

D. Fine-Grained Adaptive Learning

Fine-grained information is very important for improving the generalization performance [17], [47], [48]. Thus we introduce FAL to facilitate the adaptive learning of fine-grained forgery-aware information. We first set the weight vector corresponding to the genuine face from the last FC layer as the proxy prototype of genuine faces, as previous work [65] has demonstrated that the weights in classifier converge to the central direction of each class. In each fine-grained pair, FAL pulls closer the similarity between the prototype and the genuine face while pushing away the manipulated face from the prototype through the modified circle loss [27]:

$$L_{FAL} = \log\left[1 + \sum \exp(\eta(\gamma_n(s_n - m_n) - \gamma_p(s_p - m_p)))\right], \tag{11}$$

$$s_p = \text{CosSim}(\mathbf{F}_{real}, \mathbf{F}_{pro}), \ s_n = \text{CosSim}(\mathbf{F}_{fake}, \mathbf{F}_{pro}) \tag{12}$$

$$\gamma_p = max(1 + m - s_p, 0), \ \gamma_n = max(m + s_n, 0), \tag{13}$$

$$m_p = 1 - m, \ m_n = m, \tag{14}$$

where CosSim represents cosine similarity, and $\mathbf{F}_{pro}$ is the prototype of genuine faces. η is the scaling factor. m is a hyper-parameter to control the margin and weighting factors

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in fine-grained pairs. $\mathbf{F}_{real}$ and $\mathbf{F}_{fake}$ are encoded features of fine-grained pairs from the last ViT block.

In the optimization, the decision boundary is achieved at $\gamma_n(s_n - m_n) - \gamma_p(s_p - m_p) = 0$. By using Eqns. 13 and 14, the decision boundary is transformed to:

$$s_n^2 + (1 - s_p)^2 = 2m^2 \tag{15}$$

Eqn. 15 indicates that FAL encourages fine-grained pairs to converge towards a circle centered at $s_p = 1$ and $s_n = 0$ with a radius of $\sqrt{2}m$, as illustrated in Fig. 6. This optimization enables each fine-grained pair to provide a more flexible gradient to guide the learning of fine-grained information. For example, at point A, when $grad_f$ is much larger than $grad_r$, the model emphasizes capturing fine-grained discriminative information to push away the manipulated face. In contrast, when $grad_f$ is much smaller than $grad_r$ at point B, the model focuses on learning fine-grained consistency to pull the genuine face closer. On the other hand, FAL ignores the gradient from non-fine-grained pairs, such as the example of point C in Fig. 6. In this case, the discrepancy involves background or non-essential manipulation regions, which could potentially lead the model to overfit to trivial features.

Obviously, FAL compact genuine faces in the feature space by exploring fine-grained information in subtle discrepancy regions. It does not explicitly penalize the distance of manipulated faces, as we want to preserve the diversity of manipulation traces in different forgery technologies. Unlike Single Center Loss (SCL) [10], which uses the mean distance between different categories to compact the intra-class variance, FAL focuses on learning both fine-grained discriminative and consistent information in each fine-grained pair, making the process of mining critical information more precise.

E. Total Loss

our proposed FA-ViT is trained end-to-end and supervised by the cross-entropy loss between the prediction result $\hat{y}$ and the ground truth label y:

$$L_{ce} = -y \log \hat{y} - (1 - y) \log (1 - \hat{y}), \tag{16}$$

where the label y is 0 for genuine faces, otherwise y is 1. The overall objective function consists of two components:

$$L_{total} = L_{ce} + \lambda L_{FAL}, \tag{17}$$

where λ is a weighting parameter. In the first training epoch, we set λ to 0, allowing the model to focus on learning classification information. Subsequently, we adjust λ to 1, thereby facilitating the learning of fine-grained information.

IV. Experiments

A. Experimental Setups

1) Dataset: We adopt eight widely used public datasets to evaluate our model. 1) **FaceForensics++** (FF++) [66] is a widely used dataset which consists of four types of face manipulation techniques: DeepFakes (DF) [75], Face2Face (F2F) [76], FaceSwap (FS) [77], and NeuralTextures (NT) [78]. 2) **Celeb-DF-v2** (CDF) [68] is a high-quality Deepfake dataset specifically centered on celebrity faces. 3) **WildDeepfake** (WDF) [69] collects the Deepfake videos from the internet, which includes diverse scenes and forgery methods. The evaluation results on WDF reflect the detector's performance in real-world scenarios. 4) **Deepfake Detection Challenge** (DFDC) [71] provides a challenging dataset with a variety of Deepfake videos stemming from different scenarios, employing diverse Deepfake, GAN-based, and traditional face-swapping methods. 5) **Deepfake Detection Challenge Preview** (DFDC-P) [70] provides a preview version of DFDC dataset and incorporates two facial modification algorithms. 6) **DeepFakeDetection** (DFD) [72] serves as another comprehensive Deepfake dataset. This dataset comprises over 3,000 manipulated videos involving 28 actors and varying scenes. 7) **DeeperForenics-1.0** (DFR) [67] is created by using real videos from FF++ with an innovative end-to-end face swapping framework. It also serves as a popular dataset to measure the morel's robustness. 8) **FFIW-10K** (FFIW) [73] is a recent large-scale dataset which focuses on multi-person scenarios. For the DFD dataset, we use all the videos for evaluation. For the other datasets, we follow the official strategy to split the corresponding datasets. For a comprehensive overview, please refer to Table I.

TABLE I
THE DETAILS OF DATASETS USED FOR TRAINING AND TESTING

Dataset	Usage	Real/Fake Videos
FaceForensics++ (FF++) [66]	Train	720/2880
DeeperForensics-1.0 (DFR) [67]	Train	799/799
FaceForensics++ (FF++) [66]	Test	140/560
Celeb-DF-v2 (CDF) [68]	Test	178/340
WildDeepfake (WDF) [69]	Test	396/410
DFDC-P [70]	Test	276/504
DFDC [71]	Test	2500/2500
DeepFakeDetection (DFD) [72]	Test	363/3431
DeeperForensics-1.0 (DFR) [67]	Test	201/201
FFIW [73]	Test	2500/2500

TABLE II
THE DETAIL STRUCTURE OF CNN

Module	Layer	Output Chan.	Output Size
Head	Conv 0-1	32	112
	Conv 0-2	32	56
Block 1	Conv 1-1	64	56
	Conv 1-2	128	56
	Conv 1-3	64	56
Block 2	Conv 2-1	128	56
	Conv 2-2	256	28
	Conv 2-3	128	28
Block 3	Conv 3-1	256	28
	Conv 3-2	512	14
	Conv 3-3	256	14

2) Implementation Details: We use MTCNN [79] to crop face regions and resize them to 224×224. We sample only 20 frames from each video to build the training data. During testing, we sample 50 frames from each video for evaluation. Our method is implemented with PyTorch library [80] on a single NVIDIA GTX 3090. We adopt the ViT-Base model pre-trained on ImageNet-21K [81] as the main backbone in FA-ViT. For optimization, we employ Adam [82] optimizer with an initial learning rate of 3×10^{-5}, a weight decay of 1×10^{-5}, and a batch size of 32. The learning rate decays by 0.5 every 5 epochs. The hyper-parameters of m and η in FAL

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TABLE III
INTRA-DATASET EVALUATION ON FF++. '*' INDICATES THE TRAINED MODEL PROVIDED BY THE AUTHORS. '†' INDICATES OUR RE-IMPLEMENTATION WITH THE OFFICIAL CODE

Method	Venue	FF++ (C40)		FF++ (C23)	
		ACC	AUC	ACC	AUC
UIA-ViT [18]	ECCV 2022	-	-	-	99.33
F2Trans [17]	TIFS 2023	90.19	94.11	98.14	<u>99.73</u>
DCL* [25]	AAAI 2022	-	-	97.62	99.57
SBI* [8]	CVPR 2022	-	-	75.29	81.95
LSDA* [41]	CVPR 2024	-	-	96.67	99.42
CFM† [39]	TIFS 2023	**94.57**	**97.87**	**98.54**	99.62
RECCE† [37]	CVPR 2022	91.59	96.05	<u>98.23</u>	**99.76**
ViT† [74]	ICLR 2021	90.00	93.49	96.00	98.92
FA-ViT	Ours	<u>92.71</u>	<u>96.64</u>	97.86	99.60

will be discussed in Sec. IV-F. The detail structure of CNN is illustrated in Table II, with parameters initialized using random initialization.

3) Evaluation Metrics: We follow the evaluation strategy in [17], the Accuracy (ACC) and Area Under the receiver operating characteristic Curve (AUC) are used as our evaluation metrics. For a fair comparison, we average the predictions in the same video to obtain the video-level prediction, and the results of other works are also presented at the video-level.

B. Intra-Dataset Evaluation

We conduct intra-dataset experiments on widely used FF++ dataset, including both high-quality (C23) and low-quality (C40) datasets. All models are trained and tested on the same dataset, where the performance reveals the model's capability in capturing the manipulation traces in forged faces. Table III illustrates the intra-dataset results, where we bold and underline the best and the second-best scores, respectively.

In general, many manipulation traces are removed in C40 compressed data, making them hard to be detected. Under this challenging setting, it can be observed that our proposed FA-ViT outperforms most previous arts by a considerable margin. For example, our method surpass recent ViT-based model, F2Trans, by 2.53% in terms of AUC. When evaluating on C23 data, all SOTA methods are able to achieve very high detection ACCs. FA-ViT still achieves outstanding 97.86% ACC and 99.60% AUC scores, demonstrating its effectiveness on C23 data. Moreover, compared with the ViT baseline, FA-ViT improves the ACC from 90.00% to 92.17% under the C40 setting and 96.00% to 97.86% under the C23 setting.

Although FA-ViT achieves satisfactory detection accuracy in intra-dataset evaluations, its performance remains inferior to CFM [39] and RECCE [37]. This result may be attributed to the fact that CFM requires pixel-level labels for supervision and RECCE employs a pixel-level reconstruction task, making these methods more sensitive to intra-dataset patterns. We will investigate the possibility of other designs to further improve performance in the intra-dataset setting.

C. Cross-Dataset Evaluation

Cross-dataset evaluation is a very challenging setting because the scenarios and characters in these datasets are complex and distinct to the training data. To comprehensively evaluate the generalization performance on different unseen datasets, we conduct extensive experiments on seven benchmark datasets. Specifically, all the models are trained on FF++ (C23) and evaluated on unforeseen datasets: CDF, WDF, DFDC-P, DFDC, DFD, DFR, and FFIW.

Table IV illustrates the cross-dataset comparison results in terms of AUC. Compared to other ViT-based methods on CDF, including UIA-ViT, F2Trans, TALL-ViT and ViT, our proposed FA-ViT outperforms these methods by 4.98%, 6.37%, 7.25%, and 10.05%, respectively. Notably, CFM achieves the best intra-dataset performance, but its generalization performance is inferior to that of our proposed FA-ViT. Moreover, compared to recent methods LSDA and DSRL, our FA-ViT achieves improvements of 4.88% and 4.75% in terms of AUC, respectively, highlighting its state-of-the-art generalization capability for detecting unseen Deepfake faces. Overall, FA-ViT shows excellent performance across all seven datasets, these finds verifies the effectiveness of adaptive learning paradigm for pre-trained ViT.

D. Cross-Manipulation Evaluation

The generalization performance of face forgery detectors on new manipulation methods is very important in real-world application. We follow the protocol proposed in [17] and [25] to evaluate cross-manipulation performance, where the model is trained on various manipulation types of samples and tested on unknown manipulation methods. The results are reported in Table V. It is observed that our proposed method achieves superior performance in cross-manipulation evaluation, exceeding the recent state-of-the-art method F2Trans by 3.22% on GID-DF (C23) and 3.42% on GID-DF (C40) in terms of ACC, respectively. Notably, we observe that the fully fine-tuned ViT does not generalize well to F2F, while our proposed FA-ViT achieves substantial improvements on both GID-F2F (C23) and GID-F2F (C40), demonstrating that adaptive learning strategy greatly enhance the model's generalization performance.

E. Robustness to Real-World Perturbations

The transmission of Deepfake videos through online social networks inevitably introduces various distortions [85], such as video compression, noise, etc. To study the robustness performance on unseen perturbations, we follow the protocol proposed in DFR [67]. Specifically, diverse perturbations are applied to the test set of DFR, including single-level random-type distortions (std/sing), random-level random-type distortions (std/rand), a mixture of three random-level random-type distortions (std/mix3), and a mixture of four random-level random-type distortions (std/mix4). We train the FA-ViT and the ViT on the pristine training set of DFR.

Table VI illustrates the comparison results in terms of ACC. We can observe that other methods degrade dramatically when multiple types of perturbation are applied to pristine inputs in the std/mix3 and std/mix4 settings. For example, ViT achieves 96.77% in the std/rand setting, but its performance drops to 87.06% in the std/mix4 setting. In contrast,

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TABLE IV

CROSS-DATASET EVALUATION ON SEVEN UNSEEN DATASETS IN TERMS OF AUC. '*' INDICATES THE TRAINED MODEL PROVIDED BY THE AUTHORS. '‡' INDICATES THE RESULTS PROVIDED BY THE AUTHORS. '†' INDICATES OUR RE-IMPLEMENTATION

Method	Venue	CDF	WDF	DFDC-P	DFDC	DFD	DFR	FFIW	Average
UIA-ViT‡ [18]	ECCV 2022	88.85	-	79.54	-	-	-	-	-
F2Trans [17]	TIFS 2023	87.46	-	77.69	-	-	-	-	-
TALL-ViT [13]	ICCV 2023	86.58	-	-	74.10	-	-	-	-
DCL* [25]	AAAI 2022	88.24	77.57	76.87	75.03	92.91	97.48	86.26	85.05
SBI* [8]	CVPR 2022	88.61	70.27	84.80	71.70	82.68	89.85	86.62	82.07
LSDA* [41]	CVPR 2024	89.80	75.60	81.20	73.50	**95.60**	89.20	87.90	84.69
DSRL‡ [42]	IJCV 2024	84.56	77.31	75.64	77.99	92.32	97.23	88.69	84.82
CFM† [39]	TIFS 2023	89.65	82.27	80.22	70.59	95.21	97.59	83.81	85.62
RECCE† [37]	CVPR 2022	69.25	76.99	66.90	70.26	86.87	96.33	91.20	79.68
ViT† [74]	ICLR 2021	83.78	78.04	78.53	73.95	91.73	94.64	76.92	82.51
FA-ViT	Ours	**93.83**	**84.32**	**85.41**	**78.32**	94.88	**98.01**	**92.22**	**89.57**

TABLE V

CROSS-MANIPULATION EVALUATION ON FF++, GID-DF MEANS TRAINING ON THE OTHER THREE MANIPULATED METHODS WHILE TESTING ON DEEPFAKES CLASS. RESULTS ARE CITED FROM [17], '†' INDICATES OUR RE-IMPLEMENTATION

Method	GID-DF (C23)		GID-DF (C40)		GID-F2F (C23)		GID-F2F (C40)	
	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC
EfficientNet [2]	82.40	91.11	67.60	75.30	63.32	80.10	61.41	67.40
MLGD [83]	84.21	91.82	67.15	73.12	63.46	77.10	58.12	61.70
LTW [84]	85.60	92.70	69.15	75.60	65.60	80.20	65.70	72.40
DCL [25]	87.70	94.90	75.90	83.82	68.40	82.93	67.85	75.07
M2TR [14]	81.07	94.91	74.29	84.85	55.71	76.99	66.43	71.70
F3Net [11]	83.57	94.95	77.50	85.77	61.07	81.20	64.64	73.70
F2Trans [17]	89.64	97.47	77.86	86.92	81.43	90.55	66.79	76.52
ViT† [74]	86.79	94.18	75.14	82.09	64.29	75.40	62.38	71.12
FA-ViT	**92.86**	**98.10**	**81.28**	**87.89**	**82.57**	**91.20**	**67.91**	**76.84**

TABLE VI

ROBUST EVALUATION ON REAL-WORLD PERTURBATIONS IN TERMS OF ACC. RESULTS ARE CITED FROM [17], '†' INDICATES OUR RE-IMPLEMENTATION

Method	std (training set)			
	std/rand	std/sing	std/mix3	std/mix4
ResNet+LSTM [67]	97.13	90.63	-	-
F3Net [11]	92.79	86.82	79.10	66.42
Xception [1]	94.75	88.38	82.32	-
M2TR [14]	94.78	91.79	86.32	78.36
MTD-Net [59]	95.22	-	86.89	-
F2Trans [17]	97.26	95.77	93.28	90.55
ViT† [74]	96.77	93.53	90.05	87.06
FA-ViT	**98.76**	**96.27**	**94.03**	**91.54**

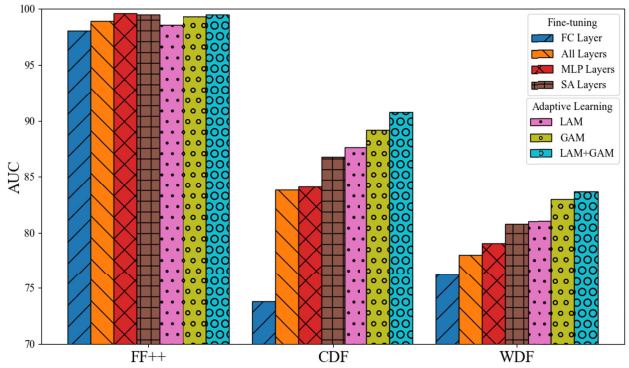


Fig. 7. The comparison between different model adaptations in terms of AUC.

TABLE VII

THE COMPARISON OF TRAINABLE PARAMETERS AND INFERENCE SPEED

Adaptation Method	#Trainable Params	#Inference Speend
FC Layer	2.30K	17.5 Iter/s
MLP Layer	28.85M	17.5 Iter/s
SA Layer	56.67M	17.5 Iter/s
All Layer	85.80M	17.5 Iter/s
LAM	12.00M	14.3 Iter/s
GAM	30.09M	13.5 Iter/s
LAM+GAM	42.09M	11.7 Iter/s

our proposed FA-ViT achieves the best performance across all distortion settings, with its performance declining more gradually in complex scenarios. The primary reason may be that we preserve the pre-trained ViT features in our adaptive learning process, as these features have demonstrated robust performance under various perturbations [86].

F. Ablation Studies

To analyze the impacts of different components in our proposed FA-ViT, we conduct ablation experiments by training all variants on FF++ (C23). We present the intra-dataset results on FF++ and cross-dataset results on CDF and WDF datasets. To ensure fair comparisons, all experiments are conducted using the same random seed.

1) Impacts of Different Model Adaptations: To validate the superiority of the proposed adaptive learning paradigm, we apply various model adaptation strategies to the same pre-trained ViT with CE loss. The results shown in Fig 7 lead to the following observations. (1) Fine-tuning only the last FC layer while keeping the pre-trained ViT fixed yields relatively high detection performance on FF++, but this strategy does not generalize well to CDF and WDF. This indicates that while pre-trained knowledge is feasible for detecting Deepfakes, it does not provide a generalized forensic representation. Therefore, it is necessary to learn additional forgery-aware information to enhance the model's expressiveness. (2) Partially fine-tuning SA layers or MLP layers achieves better generalization performance than the fully

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TABLE VIII
ABLATION ON FREEZING PRE-TRAINED PARAMETERS

Method	FF++		CDF	WDF
	ACC	AUC	AUC	AUC
w/o freezing	95.43	99.14	86.84	79.12
w/ freezing	**97.29**	**99.51**	**90.78**	**83.63**

TABLE IX
THE COMPARISON BETWEEN DIFFERENT ADAPTATIONS LAYER

Adaptation Layer	FF++		CDF	WDF
	ACC	AUC	AUC	AUC
MLP	96.14	99.07	88.35	81.78
SA	**97.86**	**99.60**	**93.83**	**84.32**

TABLE X
THE COMPARISON BETWEEN LORA AND GAM

Method	FF++		CDF	WDF
	ACC	AUC	AUC	AUC
LoRA [52]	96.82	99.21	92.05	83.36
GAM	**97.86**	**99.60**	**93.83**	**84.32**

fine-tuning strategy. However, partially fine-tuning strategy still disrupt the pre-trained feature, resulting in inferior generalization performance compared to the proposed adaptive learning strategy. (3) Adaptive learning paradigm for pre-trained ViT achieves comparatively high performance, and the best performance is observed when both LAM and GAM collaborate to capture different types of forgery-aware information. On the other hand, Table VII presents the trainable parameters and inference speeds of different model adaptation strategies. Adaptive learning methods demonstrate better performance with fewer trainable parameters compared to the fully fine-tuning strategy. Meanwhile, We observe a slight reduction in inference speed for adaptive learning methods, as additional modules are inserted into the ViT. Furthermore, as presented in Table VIII, allowing ViT parameters to update during the adaptive training process leads to significant performance drops. This demonstrates that disrupting the pre-trained features results in sub-optimal generalization performance from another perspective.

2) Ablation of GAM: While the proposed GAM is designed to be inserted into the SA layer, another option is to insert it into the MLP layer. In this case, GAM learns to capture forgery-aware information without using the self-attention mechanism, so only one residual value is generated to adapt the original output from the MLP layer. Table IX presents a comparison of inserting GAM into different layer of ViT. It can be observed that insert the proposed GAM into SA layer achieves better performance in both intra- and cross-dataset evaluation. This is because GAM is specifically designed to interact with query, key, and value tokens, allowing it to leverage the self-attention mechanism to effectively capture global forgery-aware information. Furthermore, we conduct an experiment to compare the proposed GAM with LoRA [52], which inserts the MLP-like module into the SA layer. As illustrated in Table X, the proposed GAM outperforms LoRA by 1.78% on CDF and 0.96% on WDF, indicating that using convolutional structure in GAM is more effective for face forgery detection.

TABLE XI
COMPARISON OF DIFFERENT ADAPTATION METHOD FOR SPATIAL INFORMATION IN FA-VIT. '-' MEANS WITHOUT ADAPTING SPATIAL INFORMATION TO VIT

Method	FF++		CDF	WDF
	ACC	AUC	AUC	AUC
-	97.43	99.45	90.33	82.60
Add	96.98	99.39	91.41	80.57
CA	97.57	99.55	92.31	82.15
LAM	**97.86**	**99.60**	**93.83**	**84.32**

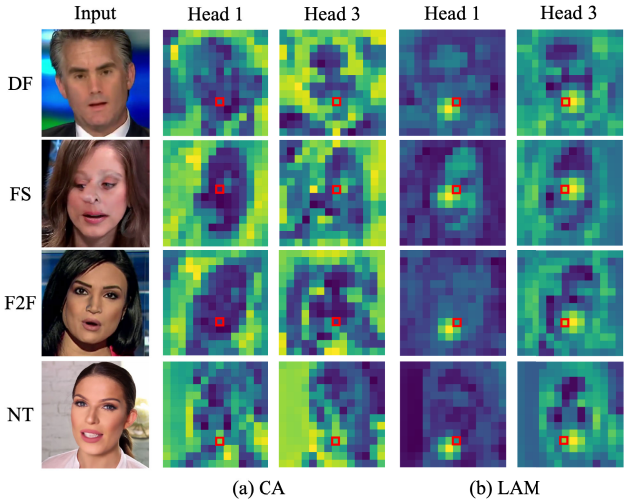
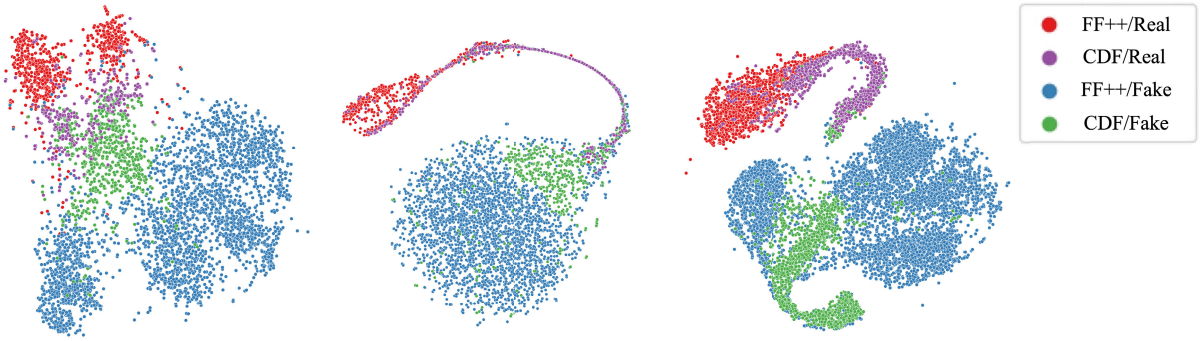


Fig. 8. Attention maps from (a) Cross Attention (CA) and (b) LAM, the query position is marked by the red box.

3) Effectiveness of LAM: To examine the effectiveness of LAM, we conduct a comparison with two commonly used methods for incorporating spatial information into ViT: (1) directly adding spatial information to each ViT token [17]; and (2) using the Cross Attention (CA) module to attentively inject spatial information into ViT [63]. As demonstrated in Table. XI, using the addition operation does not bring consistent improvements across different datasets, while CA module is not effective on WDF dataset. In contrast, LAM achieves the improvements of 3.50% on CDF and 1.72% on WDF, respectively. Unlike previous works [17], [63], where ViT parameters are allowed to collaboratively update to optimize the fusion process, adaptive learning requires the designed module to establish an effective connection between each query token and the spatial information. LAM considers the local context prior in spatial space and optimizes the combination of global and local relations in this process, thereby effectively adapting local spatial information to pre-trained ViT. Furthermore, we visualize the attention map from CA and LAM, as shown in Fig. 8. This visualization clearly shows that LAM specifically highlights the neighboring relations surrounding the query token and simultaneously interacts with long-range spatial information, demonstrating its effectiveness from another point of view.

4) Experiments on the Proposed FAL: As described in Sec. III-D, FAL is designed to expose subtle forgery discrepancies in each fine-grained pair and provides flexible optimization for model updates. We conduct the ablation experiment on FAL by designing two variations: (1) Variation 1 replaces circle loss with triplet loss; and (2) Variation

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(a) CE (b) CE + SCL (c) CE + FAL

Fig. 9. t-SNE feature distribution of our model by supervised with (a) Cross Entropy (CE) loss, (b) CE loss and Single Center Loss (SCL) [10], (c) CE loss and our proposed FAL.

TABLE XII
ABLATION EXPERIMENT ON THE PROPOSED FAL. VARIATION 1 USES TRIPLET LOSS TO REPLACE CIRCLE LOSS IN FAL, WHILE VARIANT 2 REMOVES THE CONSTRAINT THAT APPLIES ONLY TO FINE-GRAINED PAIRS

Varation	FF++		CDF	WDF
	ACC	AUC	AUC	AUC
Variation 1	97.58	99.52	92.54	83.56
Variation 2	**98.14**	**99.82**	91.28	81.71
FAL	97.86	99.60	**93.83**	**84.32**

TABLE XIII
COMPARISON EXPERIMENT ON THE PROPOSED FAL

Loss Function	FF++		CDF	WDF
	ACC	AUC	AUC	AUC
CE	97.29	99.51	90.78	83.63
CE + SCL [10]	97.43	99.46	91.67	80.60
CE + FAL	**97.86**	**99.60**	**93.83**	**84.32**

TABLE XIV
IMPACTS OF DIFFERENT COMBINATION OF m AND η IN TERMS OF AUC

Hyper-parameters		FF++	CDF	WDF
m	η			
0	24	**99.82**	90.40	81.56
0.25	24	99.60	**93.83**	**84.32**
0.5	24	99.59	92.90	82.43
0.25	1	99.44	92.10	82.48
0.25	24	99.60	**93.83**	**84.32**
0.25	32	**99.61**	92.59	83.37

TABLE XV
IMPACTS OF ZERO INITIALIZATION ON GAM AND LAM

GAM Zero Init.	LAM Zero Init.	FF++		CDF	WDF
		ACC	AUC	AUC	AUC
-	✓	97.71	99.51	88.83	80.19
✓	-	97.29	99.54	91.38	82.76
✓	✓	**97.86**	**99.60**	**93.83**	**84.32**

2 extends FAL to include non-fine-grained pairs. As the results illustrated in Table XII, Variation 1 shows inferior performance compared to the proposed FAL. This is because circle loss performs adaptive weighting for each fine-grained pair, thereby enabling the model to learn a better representation. Meanwhile, it can be observed that Variation 2 achieves slight improvements on intra-dataset evaluation but shows a heavy degradation on unseen datasets. Considering the significant differences in non-fine-grained pairs, the overfitting problem may arise from learning trivial features in non-essential regions.

Additionally, we compare the proposed FAL with SCL [10], which regularizes the mean distance between different categories to compact the intra-class variant. As illustrated in Table XIII, although SCL shows improvements on CDF, its performance on WDF decreases. In contrast, we can observe FAL achieves an improvement on CDF from 90.78% to 93.83%, and on WDF from 83.63% to 84.32%. Furthermore, the t-SNE feature distributions [87] under different optimization objects are shown in Fig. 9. Compared to SCL, the proposed FAL exhibits a more compact intra-class distribution and a more discriminative inter-class distribution, thus demonstrating its superiority in the case of adaptive learning paradigm.

5) Impacts of Different Parameters in FAL: In FAL, the parameter m controls the decision boundary's radius, while η serves as a scaling factor. Table XIV presents the intra- and cross-dataset performance under various combinations of m and η. Initially, we set η to 24 and examine the impacts of different m. We observe that setting m to 0 improves intra-dataset performance but hinders its generalization. This highlights the risk of overfitting when all genuine faces are excessively concentrated towards the center point. Consequently, we find that introducing a radius by setting $m = 0.25$ yields optimal performance. We then keep m at 0.25 and investigate the impact of η. We observe that $\eta = 1$ results in the sub-optimal performance, whereas increasing it to $\eta = 24$ produces the best results. Finally, the hyper-parameters of m and η in FAL are set as 0.25 and 24, respectively.

6) Impacts of Zero Initialization: In FA-ViT, zero initialization in GAM and LAM ensures that their outputs are zero at the beginning of training. We conduct an ablation experiment on this initialization strategy, as shown in Table XV. The results indicate that removing zero initialization leads to a decline across different evaluation settings. This is because zero initialization prevents noise from being introduced into the pre-trained features, thereby facilitating stable learning throughout the adaptive learning process.

7) Impact of Different Pre-Trained Initialization: We initialize ViT with different commonly used pre-trained weights, including DeiT [88], CLIP [89], ImageNet-1K [90],

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TABLE XVI
IMPACTS OF DIFFERENT PRE-TRAINED INITIALIZATION.'NONE' REPRESENTS THE MODEL TRAINED FROM SCRATCH

Pre-trained Initialization	FA-ViT	FF++		CDF	WDF
		ACC	AUC	AUC	AUC
None	-	70.43	74.68	60.97	57.06
None	✓	**73.57**	**80.80**	**64.23**	**57.37**
DeiT	-	**96.00**	**98.67**	82.99	80.45
DeiT	✓	95.71	98.58	**87.50**	**84.82**
CLIP	-	95.86	98.84	82.61	80.51
CLIP	✓	**97.13**	**99.33**	**90.41**	**80.81**
ImageNet-1k	-	95.86	98.85	82.76	79.92
ImageNet-1k	✓	**96.34**	**99.48**	**92.14**	**80.85**
ImageNet-21K	-	96.00	98.92	83.78	78.04
ImageNet-21K	✓	**97.86**	**99.60**	**93.83**	**84.32**

TABLE XVII
IMPACTS OF DIFFERENT VIT-BASED BACKBONES

Backbone	FA-ViT	FF++		CDF	WDF
		ACC	AUC	AUC	AUC
ViT-Small	-	94.92	98.56	81.33	77.21
ViT-Small	✓	**96.57**	**99.41**	**90.06**	**82.27**
ViT-Base	-	96.00	98.92	83.78	78.04
ViT-Base	✓	**97.86**	**99.60**	**93.83**	**84.32**
ViT-Large	-	95.43	98.93	83.44	79.74
ViT-Large	✓	**97.47**	**99.61**	**92.93**	**82.01**
T2T-ViT-14	-	95.00	98.22	80.12	80.38
T2T-ViT-14	✓	**96.57**	**99.09**	**87.92**	**82.35**
T2T-ViT-24	-	95.14	98.69	82.03	78.18
T2T-ViT-24	✓	**96.86**	**99.27**	**89.57**	**82.87**

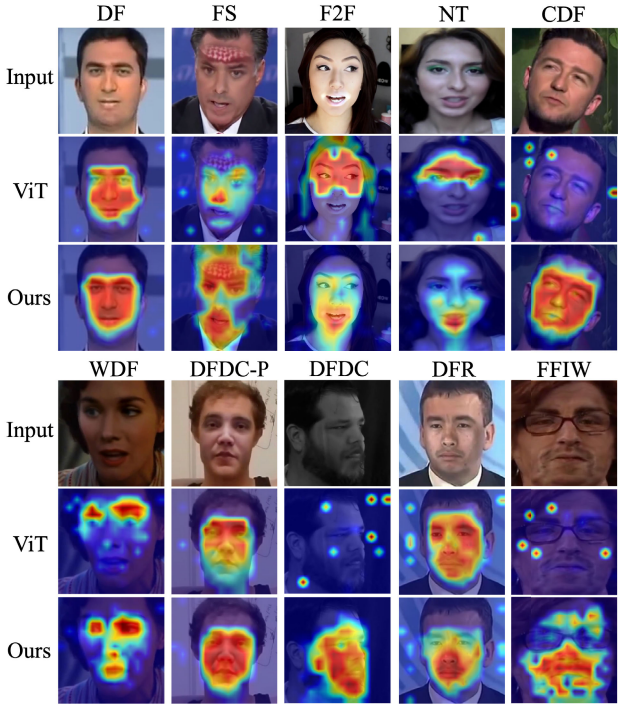


Fig. 10. Saliency map visualization of ViT and our proposed FA-ViT on different dataset.

and ImageNet-21K [81]. Additionally, ViT without pre-trained knowledge, i.e., random initialization, is evaluated as a baseline. As illustrated in Table XVI, when training from scratch, ViT-based models exhibit poor performance in both intra- and cross-dataset evaluations, indicating the importance of pre-trained knowledge in assisting model to learn forensics representations. On the other hand, FA-ViT significantly improves the generalization performance of different pre-trained ViTs compared to the fully fine-tuning strategy. We also note that different initializations can lead to certain variations in final results. Therefore, selecting an appropriate pre-trained initialization is crucial in the proposed adaptive learning process. Empirically, we find that ViT trained on ImageNet-21K shows the highest performance for face forgery detection.

8) Impacts of Different ViT-Based Backbones: Herein, we study the impacts of different ViT-based backbones, including ViT-Small, ViT-Base, ViT-Large, T2T-ViT-14 [91], and T2T-ViT-24 [91]. The results are illustrated in Table XVII. Compared with directly finetuning the vanilla ViT-based backbones, our proposed FA-ViT consistently achieves significant improvements on generalization performance across these ViT-based backbones. For example, when we apply adaptive learning to the ViT-Large backbone, the model achieves 9.49% and 2.27% AUC boosts on CDF and WDF. Similarly, FA-ViT improves the generalization performance of T2T-ViT-14 from 80.12% to 87.92% on CDF. According to this ablation experiment, we finally select ViT-Base as our basic backbone.

G. Saliency Map Visualization

To better clarify the effectiveness of our FA-ViT, we visualize the model's attention on Deepfake faces using Grad-CAM++ [92], as shown in Fig. 10. It can be observed that FA-ViT generates distinguishable saliency maps for different Deepfake faces, and captures method-specific artifacts, such as the forehead region in FS and mouth region in F2F. In cross-dataset scenarios, ViT struggles to detect Deepfakes in complex environment, such as large-pose face in CDF or challenging lighting condition in DFDC. In contrast, FA-ViT consistently traces the manipulated regions across different unseen datasets, thus verifying its effectiveness from the decision-making perspective.

V. CONCLUSION

In this paper, we present a novel Forgery-Aware Adaptive Vision Transformer (FA-ViT) tailored for generalized face forgery detection from the perspective of model adaptation, which is somewhat ignored in previous study. Specifically, the expressivity of the pre-trained ViT is preserved during training, while we design Global Adaptive Module (GAM) and Local Adaptive Module (LAM) to adapt global and local forgery-aware information for generalized representation learning. In addition, we also introduce Fine-grained Adaptive Learning (FAL) to facilitate the adaptive learning of fine-grained forgery-aware information. In conclusion, our proposed framework offers a generalized and robust solution to the challenges of face forgery detection. We believe our proposed method can provide valuable insights to the research community, and further advance the development of face forgery detection systems.

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REFERENCES

[1] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1251–1258.

[2] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6105–6114.

[3] Y. Luo, Y. Zhang, J. Yan, and W. Liu, "Generalizing face forgery detection with high-frequency features," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 16317–16326.

[4] I. Masi, A. Killekar, and R. M. Mascarenhas, "Two-branch recurrent network for isolating deepfakes in videos," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 667–684.

[5] Z. Guo, L. Wang, W. Yang, G. Yang, and K. Li, "LDFnet: Lightweight dynamic fusion network for face forgery detection by integrating local artifacts and global texture information," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 2, pp. 1255–1265, Feb. 2023.

[6] L. Li et al., "Face X-ray for more general face forgery detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5001–5010.

[7] D. Nguyen et al., "LAA-net: Localized artifact attention network for quality-agnostic and generalizable deepfake detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 17395–17405.

[8] K. Shiohara and T. Yamasaki, "Detecting deepfakes with self-blended images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 18699–18708.

[9] K. Xu, M. Qin, F. Sun, Y. Wang, Y.-K. Chen, and F. Ren, "Learning in the frequency domain," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 1740–1749.

[10] J. Li, H. Xie, J. Li, Z. Wang, and Y. Zhang, "Frequency-aware discriminative feature learning supervised by single-center loss for face forgery detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6458–6467.

[11] Y. Qian, G. Yin, L. Sheng, Z. Chen, and J. Shao, "Thinking in frequency: Face forgery detection by mining frequency-aware clues," in *Proc. 16th Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 86–103.

[12] Z. Liu, X. Qi, and P. H. S. Torr, "Global texture enhancement for fake face detection in the wild," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8060–8069.

[13] Y. Xu, J. Liang, G. Jia, Z. Yang, Y. Zhang, and R. He, "TALL: Thumbnail layout for deepfake video detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 22601–22611.

[14] J. Wang et al., "M2TR: Multi-modal multi-scale transformers for deepfake detection," in *Proc. Int. Conf. Multimedia Retr.*, Jun. 2022, pp. 615–623.

[15] J. Liu, J. Wang, P. Zhang, C. Wang, D. Xie, and S. Pu, "Multi-scale wavelet transformer for face forgery detection," in *Proc. Asian Conf. Comput. Vis.*, Jan. 2023, pp. 1858–1874.

[16] C. Zhao, C. Wang, G. Hu, H. Chen, C. Liu, and J. Tang, "ISTVT: Interpretable spatial–temporal video transformer for deepfake detection," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 1335–1348, 2023.

[17] C. Miao, Z. Tan, Q. Chu, H. Liu, H. Hu, and N. Yu, "F^2Trans: High-frequency fine-grained transformer for face forgery detection," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 1039–1051, 2023.

[18] W. Zhuang et al., "UIA-ViT: Unsupervised inconsistency-aware method based on vision transformer for face forgery detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2022, pp. 391–407.

[19] C. Miao, Z. Tan, Q. Chu, N. Yu, and G. Guo, "Hierarchical frequency-assisted interactive networks for face manipulation detection," *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 3008–3021, 2022.

[20] K. Nakashima, H. Kataoka, A. Matsumoto, K. Iwata, N. Inoue, and Y. Satoh, "Can vision transformers learn without natural images?" in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2022, vol. 36, no. 2, pp. 1990–1998.

[21] U. Ojha, Y. Li, and Y. J. Lee, "Towards universal fake image detectors that generalize across generative models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 24480–24489.

[22] A. Kumar, A. Raghunathan, R. Jones, T. Ma, and P. Liang, "Fine-tuning can distort pretrained features and underperform out-of-distribution," in *Proc. Int. Conf. Learn. Represent.*, Jan. 2022, pp. 1–54. [Online]. Available: https://openreview.net/forum?id=UYneFzXSJWh

[23] S. Chen et al., "AdaptFormer: Adapting vision transformers for scalable visual recognition," in *Proc. NIPS*, 2022, pp. 16664–16678.

[24] J.-B. Cordonnier, A. Loukas, and M. Jaggi, "On the relationship between self-attention and convolutional layers," in *Proc. Int. Conf. Learn. Represent.*, Jan. 2019, pp. 1–13. [Online]. Available: https://openreview.net/forum?id=HJlnC1rKPB

[25] K. Sun, T. Yao, S. Chen, S. Ding, J. Li, and R. Ji, "Dual contrastive learning for general face forgery detection," in *Proc. AAAI Conf. Artif. Intell.*, 2022, vol. 36, no. 2, pp. 2316–2324.

[26] W. Lu et al., "Detection of deepfake videos using long-distance attention," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 9366–9379, Jul. 2023.

[27] Y. Sun et al., "Circle loss: A unified perspective of pair similarity optimization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jul. 2020, pp. 6398–6407.

[28] F. Matern, C. Riess, and M. Stamminger, "Exploiting visual artifacts to expose deepfakes and face manipulations," in *Proc. IEEE Winter Appl. Comput. Vis. Workshops (WACVW)*, Jan. 2019, pp. 83–92.

[29] Y. Li and S. Lyu, "Exposing DeepFake videos by detecting face warping artifacts," 2018, *arXiv:1811.00656*.

[30] Y. Li, M. Chang, and S. Lyu, "In ictu oculi: Exposing AI created fake videos by detecting eye blinking," in *Proc. IEEE Int. Workshop Inf. Forensics Secur. (WIFS)*, Dec. 2018, pp. 1–7.

[31] D. Afchar, V. Nozick, J. Yamagishi, and I. Echizen, "MesoNet: A compact facial video forgery detection network," in *Proc. IEEE Int. Workshop Inf. Forensics Secur. (WIFS)*, Dec. 2018, pp. 1–7.

[32] H. H. Nguyen, J. Yamagishi, and I. Echizen, "Capsule-forensics: Using capsule networks to detect forged images and videos," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2019, pp. 2307–2311.

[33] Z. Yin et al., "Improving deepfake detection generalization by invariant risk minimization," *IEEE Trans. Multimedia*, vol. 26, pp. 6785–6798, 2024.

[34] J. Hu, X. Liao, W. Wang, and Z. Qin, "Detecting compressed deepfake videos in social networks using frame-temporality two-stream convolutional network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 3, pp. 1089–1102, Mar. 2022.

[35] X. Li, R. Ni, P. Yang, Z. Fu, and Y. Zhao, "Artifacts-disentangled adversarial learning for deepfake detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 4, pp. 1658–1670, Apr. 2022.

[36] D. Zhang, J. Chen, X. Liao, F. Li, J. Chen, and G. Yang, "Face forgery detection via multi-feature fusion and local enhancement," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 9, pp. 8972–8977, Sep. 2024.

[37] J. Cao, C. Ma, T. Yao, S. Chen, S. Ding, and X. Yang, "End-to-end reconstruction-classification learning for face forgery detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4113–4122.

[38] J. Wang, Y. Sun, and J. Tang, "LiSiam: Localization invariance Siamese network for deepfake detection," *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 2425–2436, 2022.

[39] A. Luo, C. Kong, J. Huang, Y. Hu, X. Kang, and A. C. Kot, "Beyond the prior forgery knowledge: Mining critical clues for general face forgery detection," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 1168–1182, 2023.

[40] H. Liu et al., "Spatial-phase shallow learning: Rethinking face forgery detection in frequency domain," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 772–781.

[41] Z. Yan, Y. Luo, S. Lyu, Q. Liu, and B. Wu, "Transcending forgery specificity with latent space augmentation for generalizable deepfake detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 8984–8994.

[42] J. Cao, K.-Y. Zhang, T. Yao, S. Ding, X. Yang, and C. Ma, "Towards unified defense for face forgery and spoofing attacks via dual space reconstruction learning," *Int. J. Comput. Vis.*, vol. 132, pp. 5862–5887, Jul. 2024.

[43] M. Naseer, K. Ranasinghe, S. Khan, M. Hayat, F. Khan, and M.-H. Yang, "Intriguing properties of vision transformers," in *Proc. Adv. Neural Inf. Process. Syst.*, A. Beygelzimer, Y. Dauphin, P. Liang, and J. W. Vaughan, Eds., 2021, pp. 23296–23308. [Online]. Available: https://openreview.net/forum?id=o2mbl-Hmfgd

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[44] Z. Peng et al., "Conformer: Local features coupling global representations for visual recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 367–376.

[45] W. Bai, Y. Liu, Z. Zhang, B. Li, and W. Hu, "AUNet: Learning relations between action units for face forgery detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 24709–24719.

[46] Y. Zhang, B. Colman, X. S. Guo, A. Shahriyari, and G. Bharaj, "Common sense reasoning for deepfake detection," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Nov. 2024, pp. 399–415.

[47] H. Zhao, T. Wei, W. Zhou, W. Zhang, D. Chen, and N. Yu, "Multi-attentional deepfake detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2185–2194.

[48] Q. Gu, S. Chen, T. Yao, Y. Chen, S. Ding, and R. Yi, "Exploiting fine-grained face forgery clues via progressive enhancement learning," in *Proc. AAAI Conf. Artif. Intell.*, 2022, vol. 36, no. 1, pp. 735–743.

[49] Y. Lee et al., "Surgical fine-tuning improves adaptation to distribution shifts," in *Proc. 11th Int. Conf. Learn. Represent.*, Jan. 2022, pp. 1–12. [Online]. Available: https://openreview.net/forum?id=APuPRxjHvZ

[50] J. O. Zhang, A. Sax, A. Zamir, L. Guibas, and J. Malik, "Side-tuning: A baseline for network adaptation via additive side networks," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*, Glasgow, U.K. Cham, Switzerland: Springer, 2020, pp. 698–714.

[51] J. Pan, Z. Lin, X. Zhu, J. Shao, and H. Li, "ST-adapter: Parameter-efficient Image-to-Video transfer learning," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2022, pp. 26462–26477.

[52] J. E. Hu et al., "LoRA: Low-rank adaptation of large language models," in *Proc. Int. Conf. Learn. Represent.*, Jan. 2021, pp. 1–16. [Online]. Available: https://openreview.net/forum?id=nZeVKeeFYf9

[53] M. Xu, Z. Zhang, F. Wei, H. Hu, and X. Bai, "Side adapter network for open-vocabulary semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 2945–2954.

[54] H.-P. Huang et al., "Adaptive transformers for robust few-shot cross-domain face anti-spoofing," in *Proc. 17th Eur. Conf.*, 2022, pp. 37–54.

[55] X. Ma et al., "HDDA: Human-perception-centric deepfake detection adapter," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jun. 2024, pp. 1–9.

[56] R. Shao, T. Wu, L. Nie, and Z. Liu, "DeepFake-adapter: Dual-level adapter for DeepFake detection," *Int. J. Comput. Vis.*, 2024.

[57] Y. Wang, K. Yu, C. Chen, X. Hu, and S. Peng, "Dynamic graph learning with content-guided spatial-frequency relation reasoning for deepfake detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7278–7287.

[58] Y. Wang, C. Chen, N. Zhang, and X. Hu, "WATCHER: Wavelet-guided texture-content hierarchical relation learning for deepfake detection," *Int. J. Comput. Vis.*, vol. 132, no. 10, pp. 4746–4767, Oct. 2024.

[59] J. Yang, A. Li, S. Xiao, W. Lu, and X. Gao, "MTD-Net: Learning to detect deepfakes images by multi-scale texture difference," *IEEE Trans. Inf. Forensics Security*, vol. 16, pp. 4234–4245, 2021.

[60] Z. Yu et al., "Searching central difference convolutional networks for face anti-spoofing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5295–5305.

[61] S. Chen, T. Yao, Y. Chen, S. Ding, J. Li, and R. Ji, "Local relation learning for face forgery detection," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 2, pp. 1081–1088.

[62] J. Fei, Y. Dai, P. Yu, T. Shen, Z. Xia, and J. Weng, "Learning second order local anomaly for general face forgery detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 20238–20248.

[63] L. Song, X. Li, Z. Fang, Z. Jin, Y. Chen, and C. Xu, "Face forgery detection via symmetric transformer," in *Proc. 30th ACM Int. Conf. Multimedia*, Oct. 2022, pp. 4102–4111.

[64] J. Guan et al., "Delving into sequential patches for deepfake detection," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, Jan. 2022, pp. 4517–4530.

[65] Y. Liu, G. Song, J. Shao, X. Jin, and X. Wang, "Transductive centroid projection for semi-supervised large-scale recognition," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 70–86.

[66] A. Rossler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Niessner, "FaceForensics++: Learning to detect manipulated facial images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1–11.

[67] L. Jiang, R. Li, W. Wu, C. Qian, and C. C. Loy, "DeeperForensics-1.0: A large-scale dataset for real-world face forgery detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 2889–2898.

[68] Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu, "Celeb-DF: A large-scale challenging dataset for DeepFake forensics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3207–3216.

[69] B. Zi, M. Chang, J. Chen, X. Ma, and Y.-G. Jiang, "WildDeepfake: A challenging real-world dataset for deepfake detection," in *Proc. 28th ACM Int. Conf. Multimedia*, Oct. 2020, pp. 2382–2390.

[70] B. Dolhansky, R. Howes, B. Pflaum, N. Baram, and C. Canton Ferrer, "The deepfake detection challenge (DFDC) preview dataset," 2019, *arXiv:1910.08854*.

[71] B. Dolhansky et al., "The deepfake detection challenge (DFDC) dataset," 2020, *arXiv:2006.07397*.

[72] (2019). *Dfd Dataset*. [Online]. Available: https://ai.googleblog.com/2019/09/contributing-data-to-deepfake-detection.html

[73] T. Zhou, W. Wang, Z. Liang, and J. Shen, "Face forensics in the wild," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 5778–5788.

[74] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent.*, Jan. 2020, pp. 1–11. [Online]. Available: https://openreview.net/forum?id=YicbFdNTTy

[75] (2018). *Github Deepfake Faceswap*. [Online]. Available: https://github.com/deepfakes/faceswap

[76] (2019). *Deepfakes Faceswap*. [Online]. Available: https://github.com/deepfakes/faceswap

[77] J. Thies, M. Zollhöfer, M. Stamminger, C. Theobalt, and M. Nießner, "Face2Face: Real-time face capture and reenactment of RGB videos," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2387–2395.

[78] J. Thies, M. Zollhöfer, and M. Nießner, "Deferred neural rendering: Image synthesis using neural textures," *ACM Trans. Graph.*, vol. 38, no. 4, pp. 1–12, Aug. 2019.

[79] K. Zhang, Z. Zhang, Z. Li, and Y. Qiao, "Joint face detection and alignment using multitask cascaded convolutional networks," *IEEE Signal Process. Lett.*, vol. 23, no. 10, pp. 1499–1503, Oct. 2016.

[80] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2019, pp. 1–17.

[81] T. Ridnik, E. Ben-Baruch, A. Noy, and L. Zelnik-Manor, "ImageNet-21K pretraining for the masses," 2021, *arXiv:2104.10972*.

[82] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[83] D. Li, Y. Yang, Y.-Z. Song, and T. M. Hospedales, "Learning to generalize: Meta-learning for domain generalization," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2018, vol. 32, no. 1, pp. 1–8.

[84] K. Sun et al., "Domain general face forgery detection by learning to weight," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, 2021, pp. 2638–2646.

[85] H. Wu, J. Zhou, J. Tian, J. Liu, and Y. Qiao, "Robust image forgery detection against transmission over online social networks," *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 443–456, 2022.

[86] Y. Li and C. Xu, "Trade-off between robustness and accuracy of vision transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7558–7568.

[87] L. V. D. Maaten and G. E. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 86, pp. 2579–2605, Jan. 2008.

[88] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jegou, "Training data-efficient image transformers & distillation through attention," in *Proc. Int. Conf. Mach. Learn.*, Jul. 2021, pp. 10347–10357.

[89] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.

[90] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Sep. 2009, pp. 248–255.

[91] L. Yuan et al., "Tokens-to-token ViT: Training vision transformers from scratch on imagenet," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2021, pp. 558–567.

[92] A. Chattopadhay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2018, pp. 839–847.

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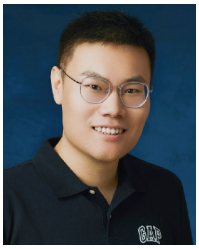
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